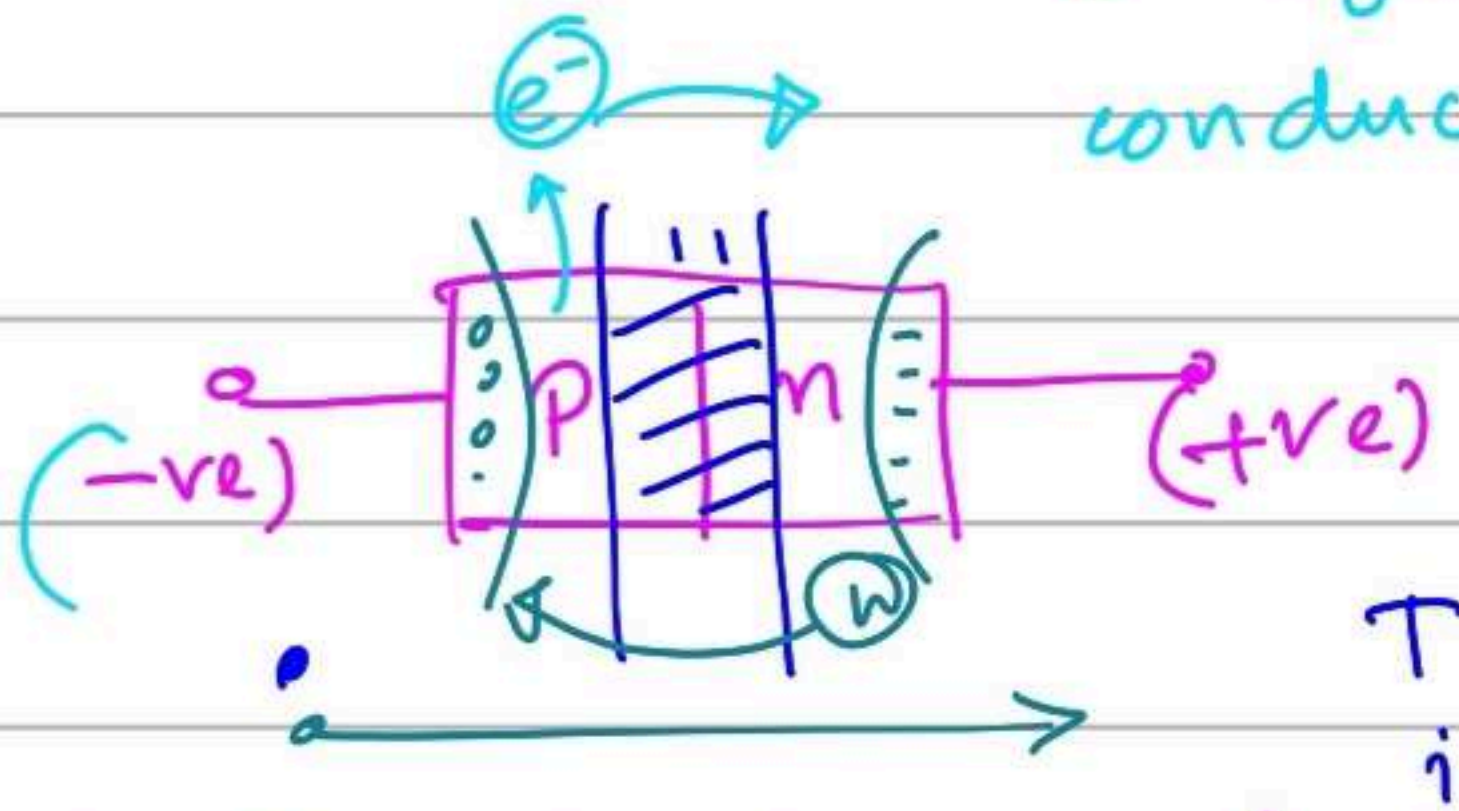


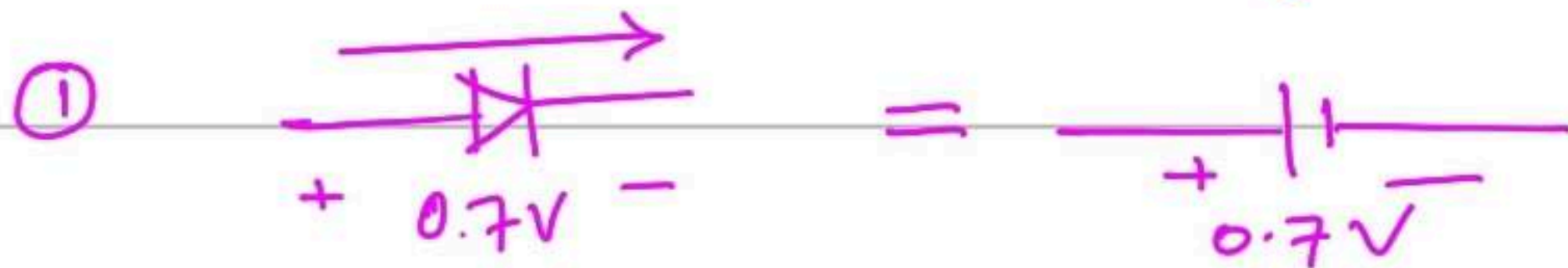
Breakdown:- If the applied reverse bias voltage is high then diode might start conduction.



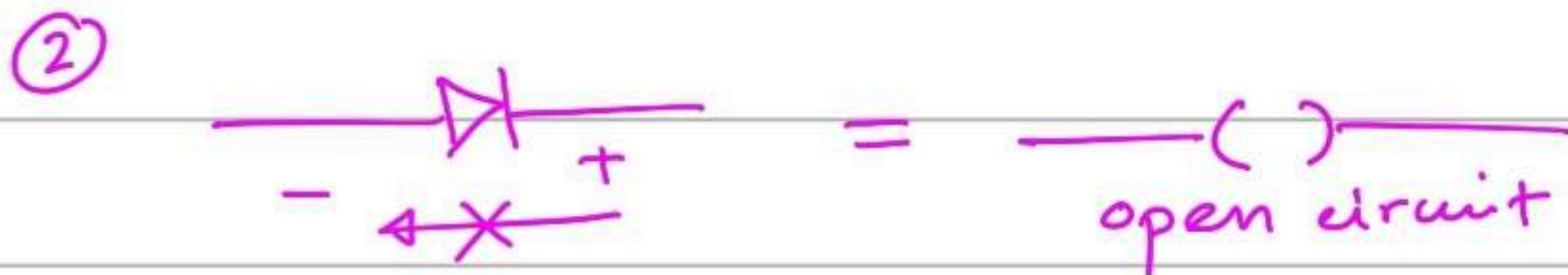
typical Si diode 25V

Zener diode $\sim 5V, 10V$

Diode $\rightarrow$ constant voltage drop model

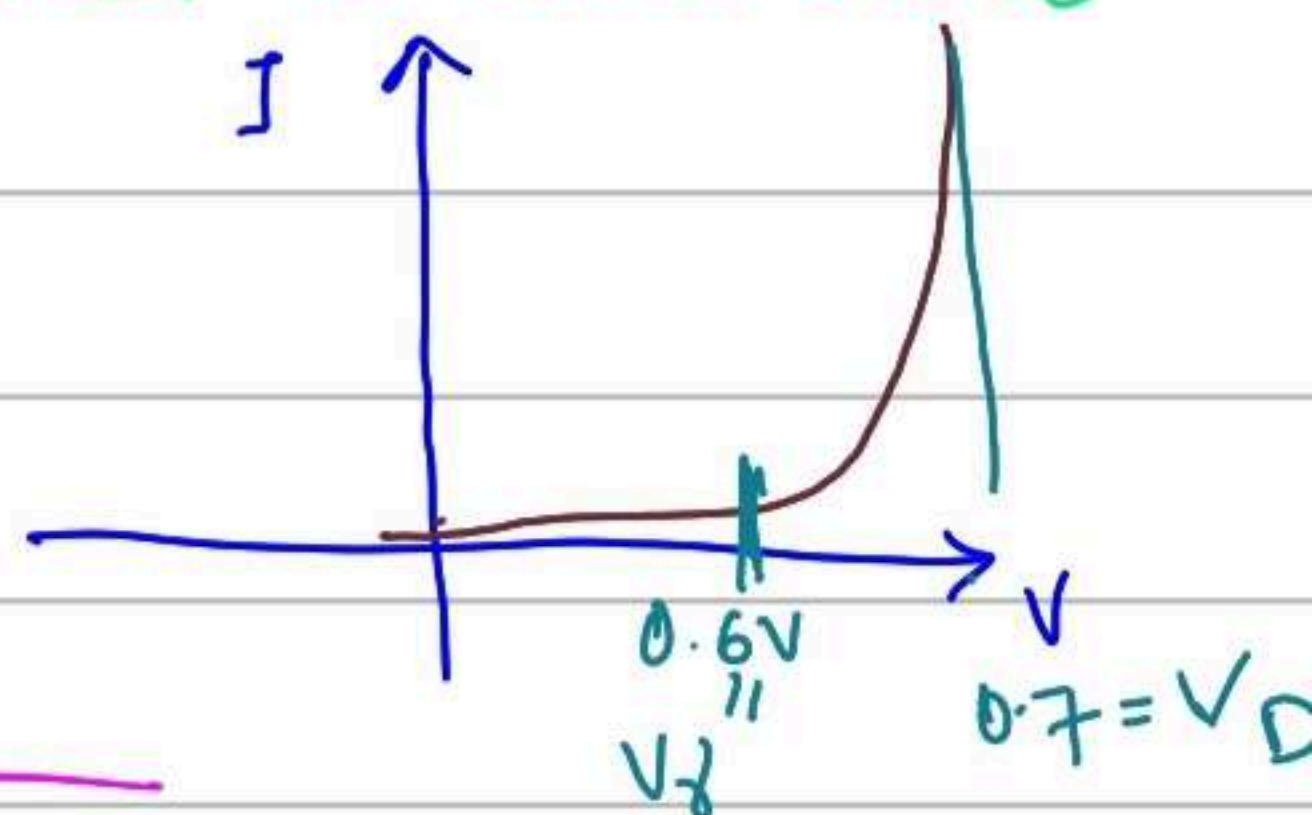


Diode conduction voltage, $V_D = 0.7V$

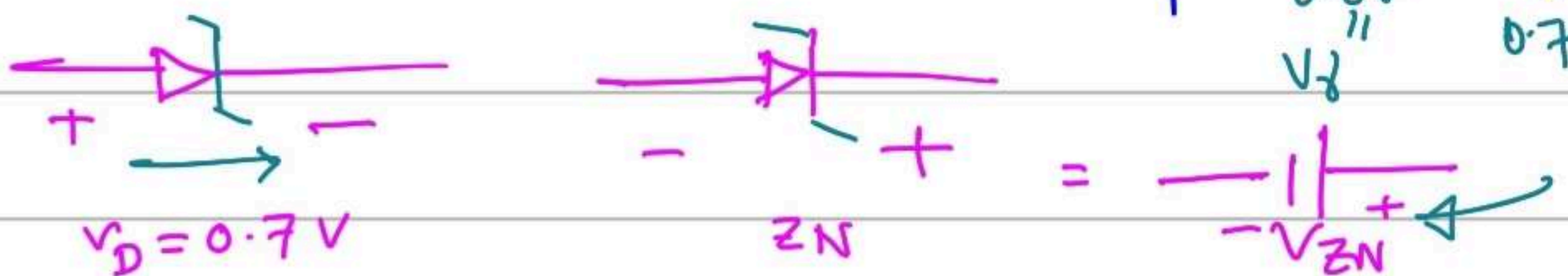


③ Activation/Threshold/Turn on voltage:

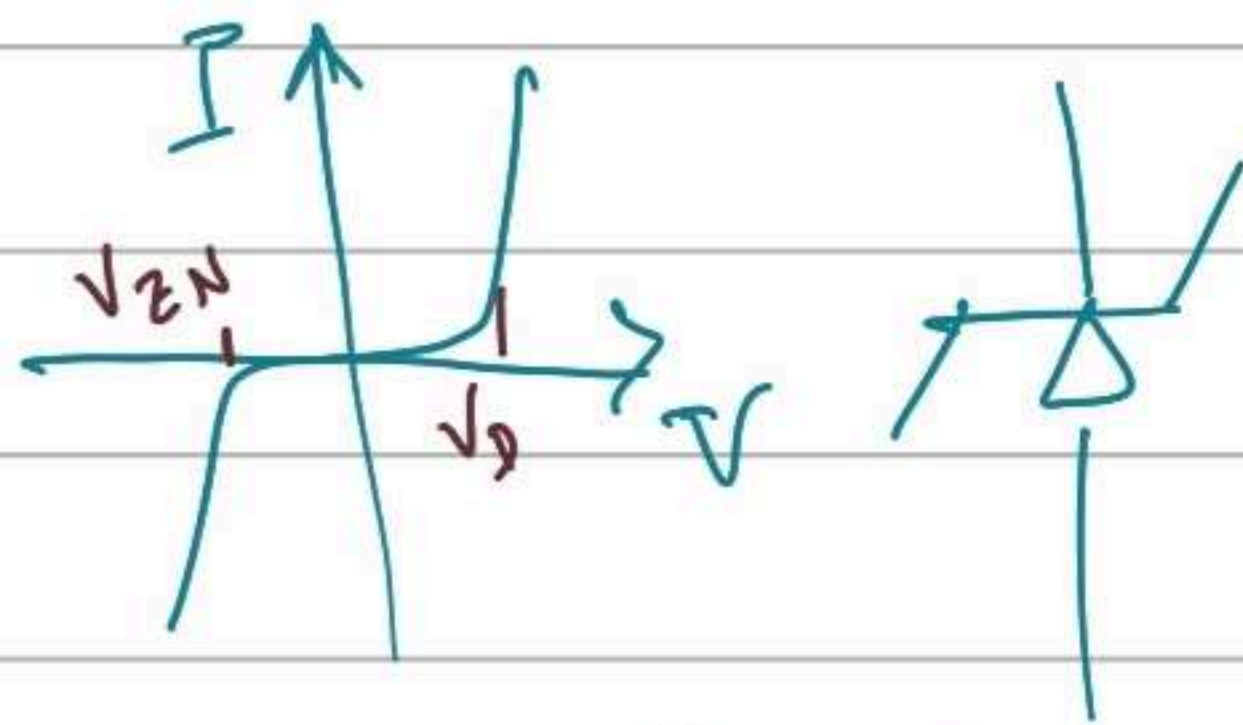
The minimum voltage required to turn on a diode. $V_T(\text{diode}) = 0.6V$



④ Zener Diode:



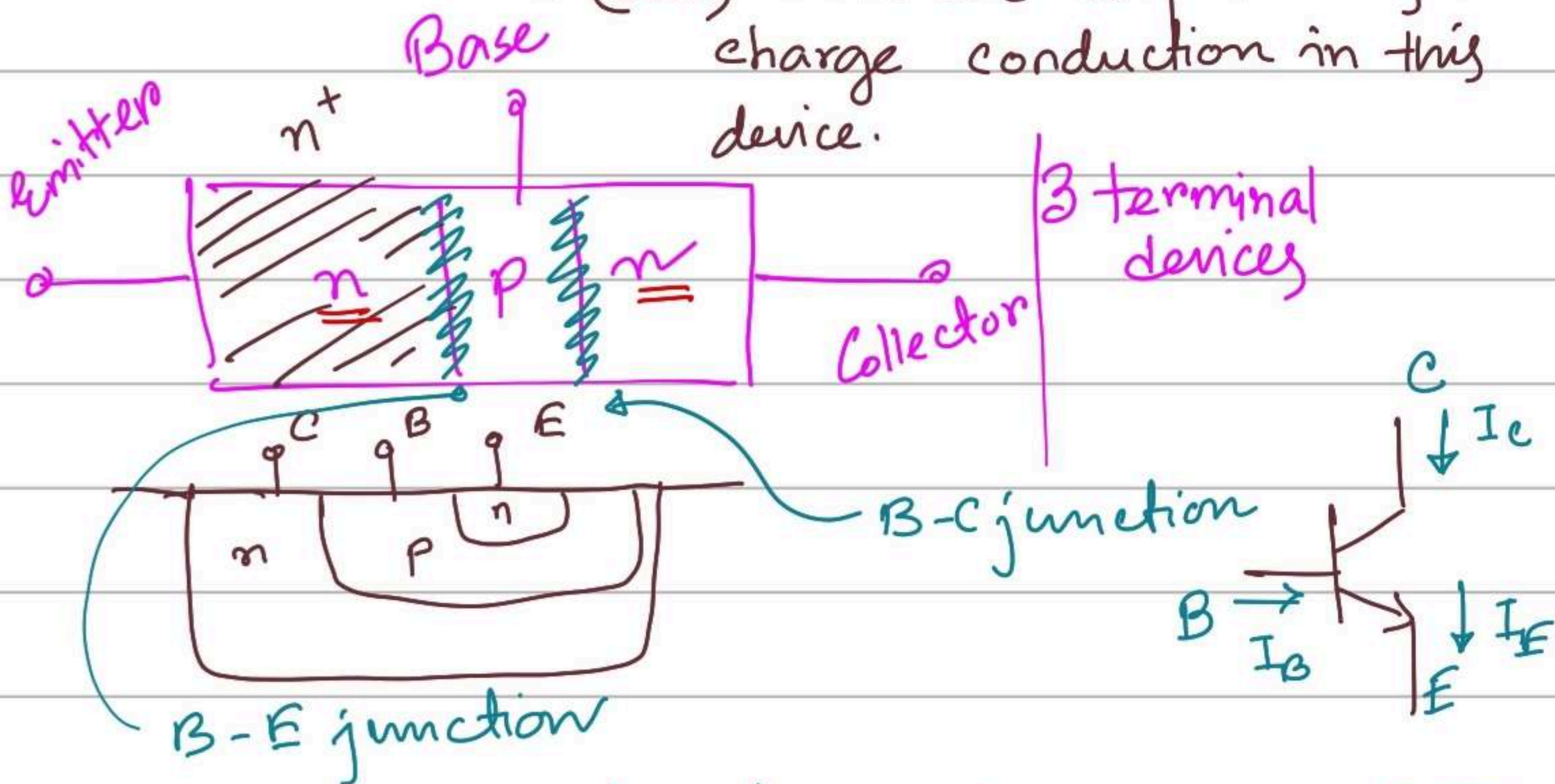
If the applied reverse bias voltage is higher than V_{ZN} , Zener diode starts conducting.



Transistor: Switching Device (~~Amplification~~)

n-p-n | Bipolar Junction Transistor (BJT)

(e-n) both are responsible for charge conduction in this device.



Depending junctions' operating mode BJT's operating mode can be classified into four different categories.

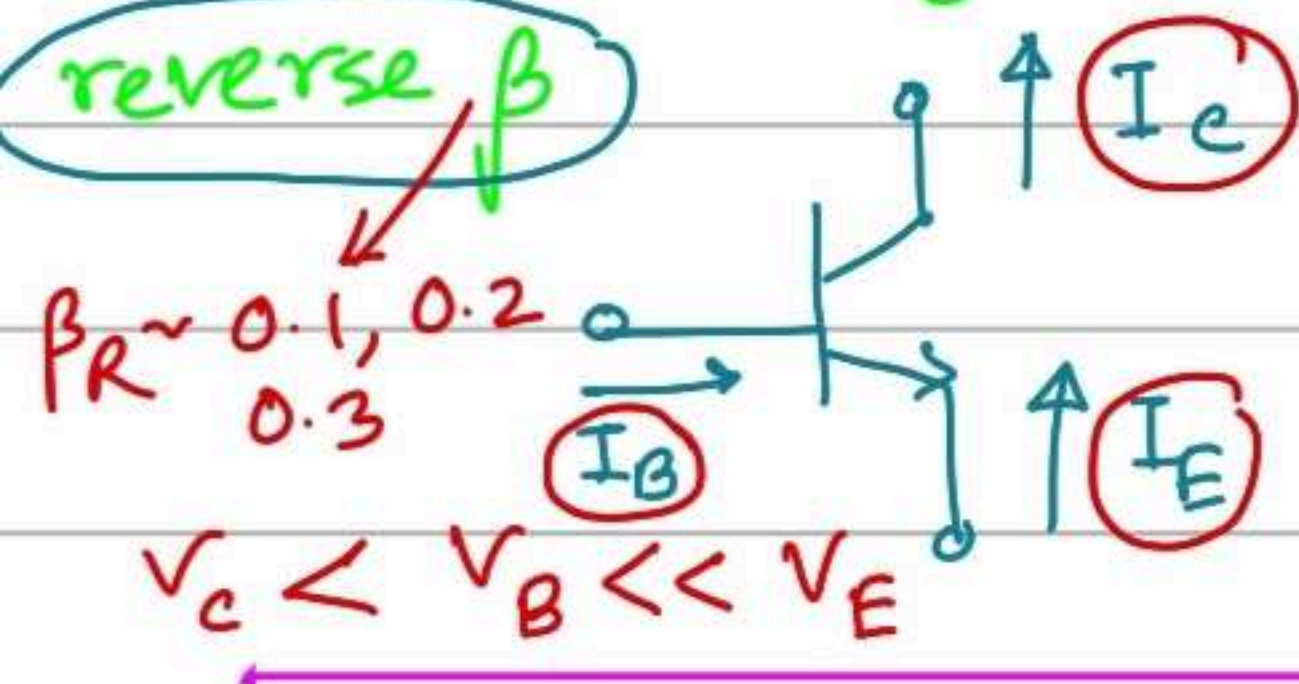
B-C junction	B-E junction	mode
Reverse	Forward	Forward Active (FA)
<u>Forward</u>	<u>Forward</u>	<u>Saturation (S)</u>
Reverse	Reverse	Cutoff (C)
<u>Forward</u>	<u>Reverse</u>	<u>Reverse Active (RA)</u>

Reverse Active:

Emitter and Collector switch their roles.

$$V_{BC} = 0.7V, \quad \frac{I_E}{I_B} = \beta_R$$

reverse β



Cutoff: ($V_{BE} < 0, V_{BC} < 0$)

$V_{BE} < V_\gamma$ (transistor)
= cut-in voltage / turn on voltage = 0.5V

$$I_C = I_B = I_E = 0$$

Checking condition:

$$\frac{I_C}{I_B} = \beta_{\text{forced}} < \beta_F (\text{Forward})$$

Forward Active

$V_{BE} > 0, V_{BC} < 0$

$$V_{BE} = 0.7, \quad \frac{I_C}{I_B} = \beta_F$$

Forward β , common emitter current gain

Checking condition:-

$$V_{CE} > 0.2V$$

$$I_C + I_B = I_E$$

Lecture 3: Introduction to Digital Logic Families

Digital logic design $\rightarrow$ { INVERTER, NAND, NOR }

Digital logic family are different groups of electronic circuits that can be used to construct digital logic gates.

Digital Logic Family

Bipolar logic family

(electron and hole both contribute in current conduction)

B.J.T.

Unipolar logic family

(only single type of carrier contributes in current conduction)
electron or hole.

JFET, MOSFET etc.

Saturated logic family

(main transistors will operate in saturation and cutoff)

- (a) Resistor Transistor Logic (RTL) ✓
- (b) Diode Transistor Logic (DTL) ✓
- (c) Direct Coupled Transistor Logic (DCTL)
- (d) Integrated Injection Logic (IIL)
- (e) High Threshold Logic (HTL) ✓
- (f) Transistor transistor Logic (TTL) ✓

Unsaturated logic family

(main transistors will operate in forward active and cutoff)

- ✓ (i) PMOS logic family
- ✓ (ii) NMOS ..
- ✓ (iii) CMOS ..

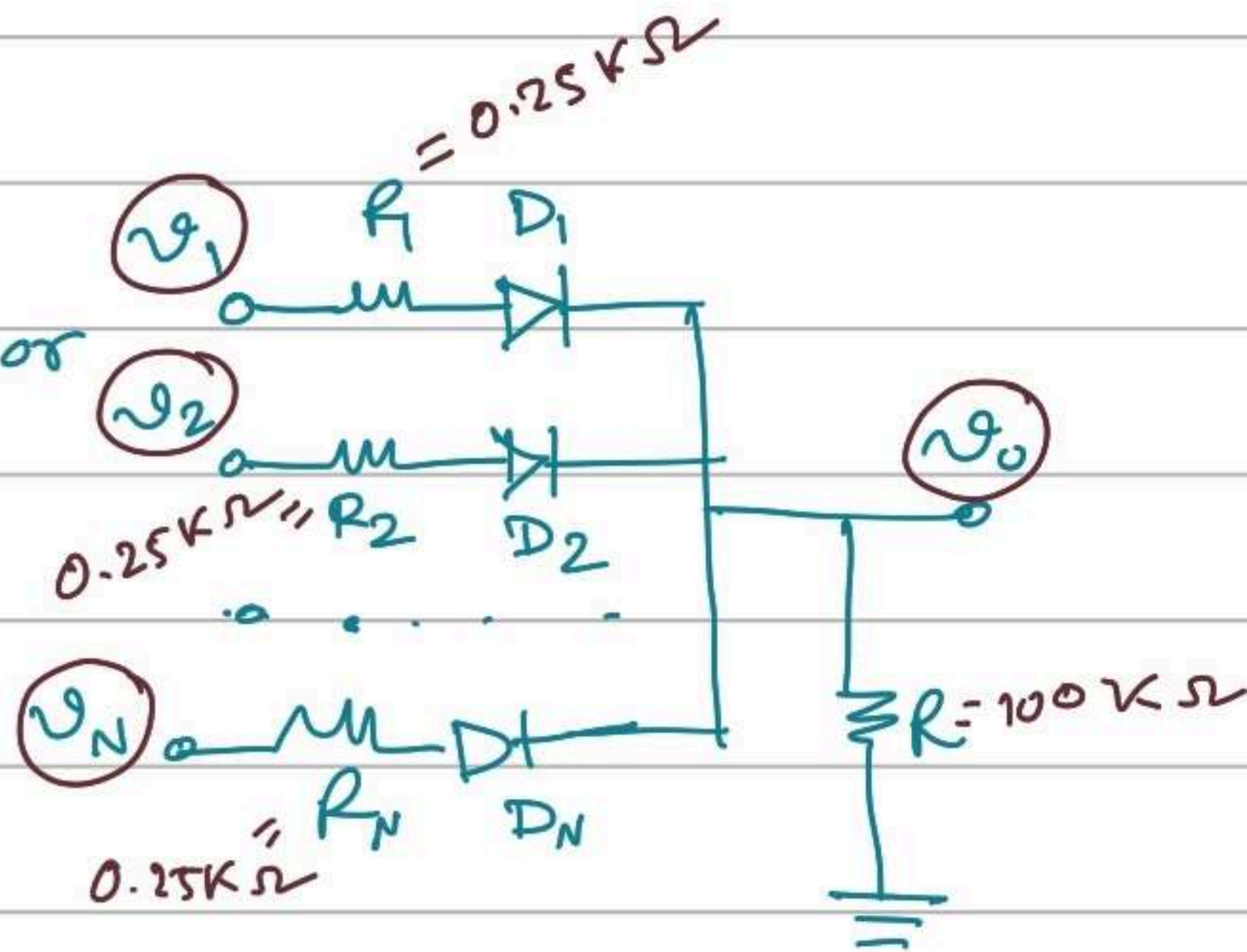
(a) Schottky TTL

(b) Emitter Coupled Logic (ECL) ✓

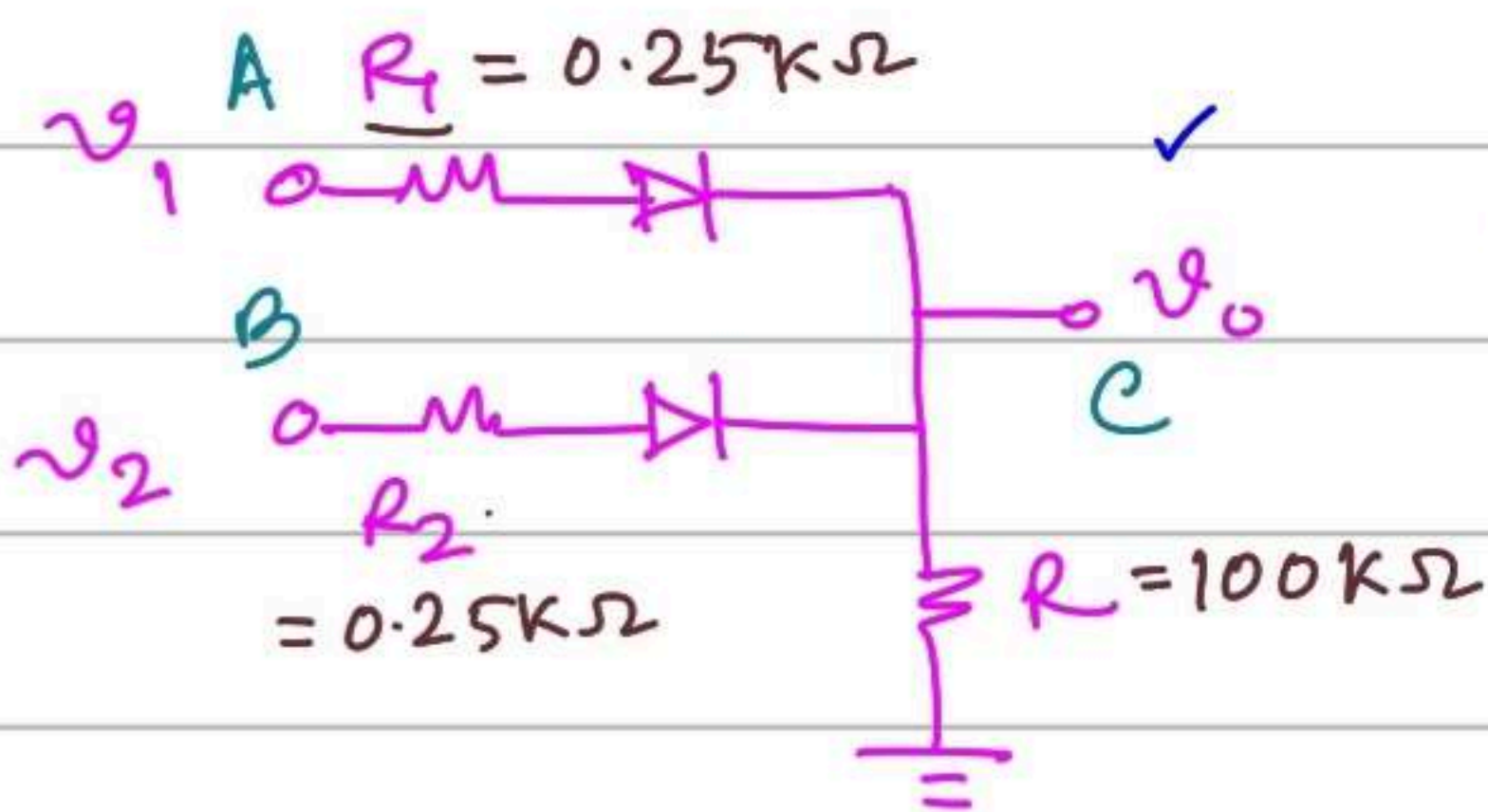
Diode Logic :-

OR GATE :-

N number of inputs for OR gate



Operation :- 2-OR GATE

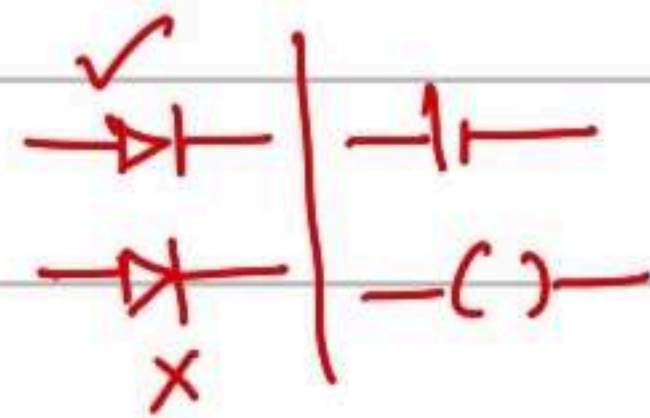
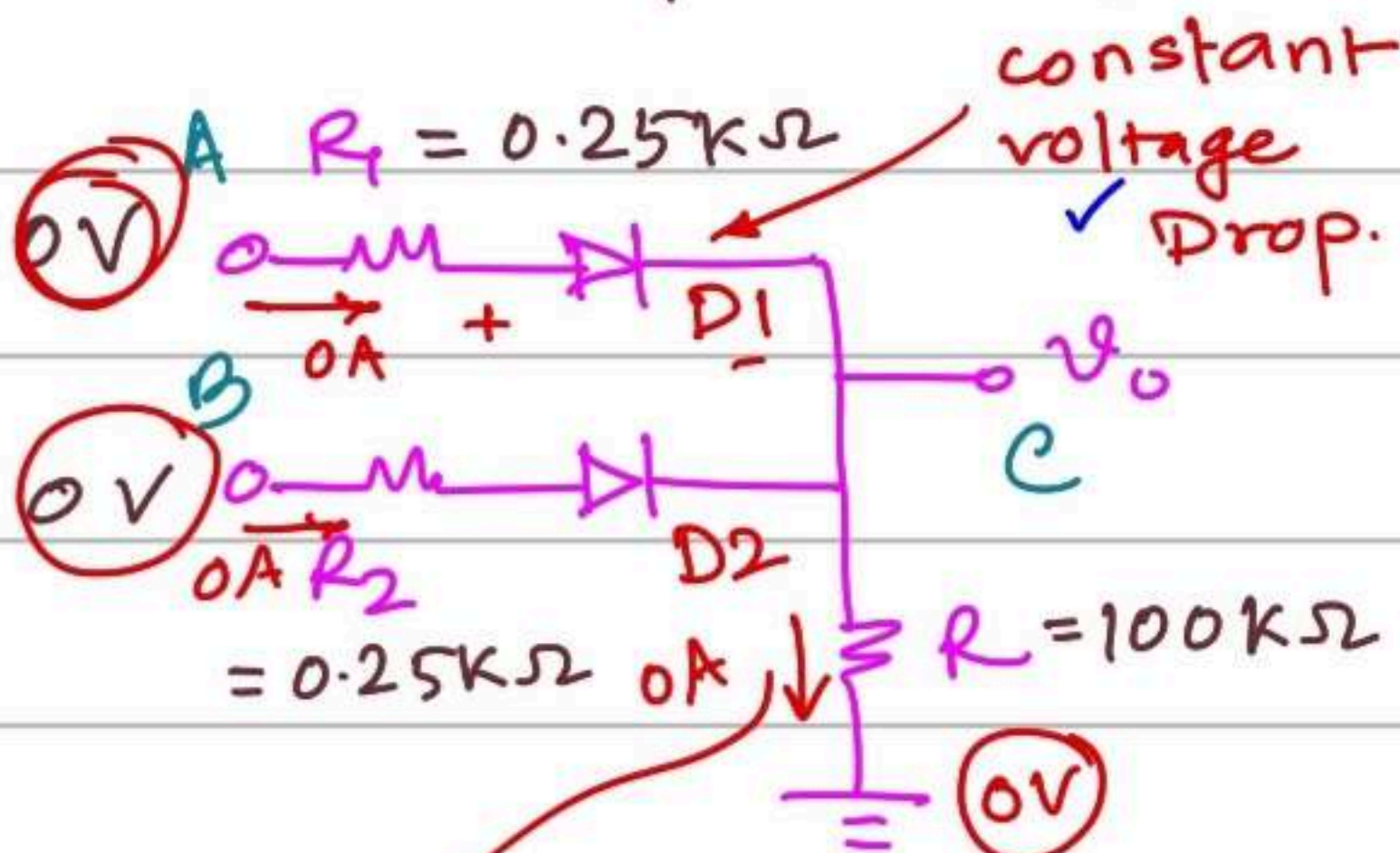


	A	B	v_1	v_2	v_0	C
①	0	0	0V	0V	0V	0
②	0	1	0V	5V	4.289	1
③	1	0	5V	0V	4.289	1
④	1	1	5V	5V	4.294	1

We will assume logic 1 for input part is 5V,
and logic 0 for input part is 0V.

Case 1:

$$v_1 = 0V, v_2 = 0V$$



There is no higher voltage input terminal in this circuit. Thus diode will never turn on.

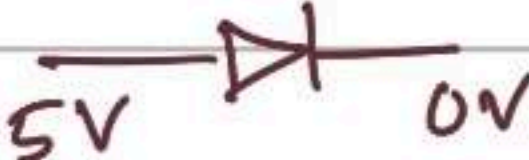
$$i = \frac{v_0 - 0}{R} \Rightarrow 0 = \frac{v_0}{R} \therefore v_0 = 0V$$

Case (2) & (3)

Case (a)

* D1, OFF, D2, OFF

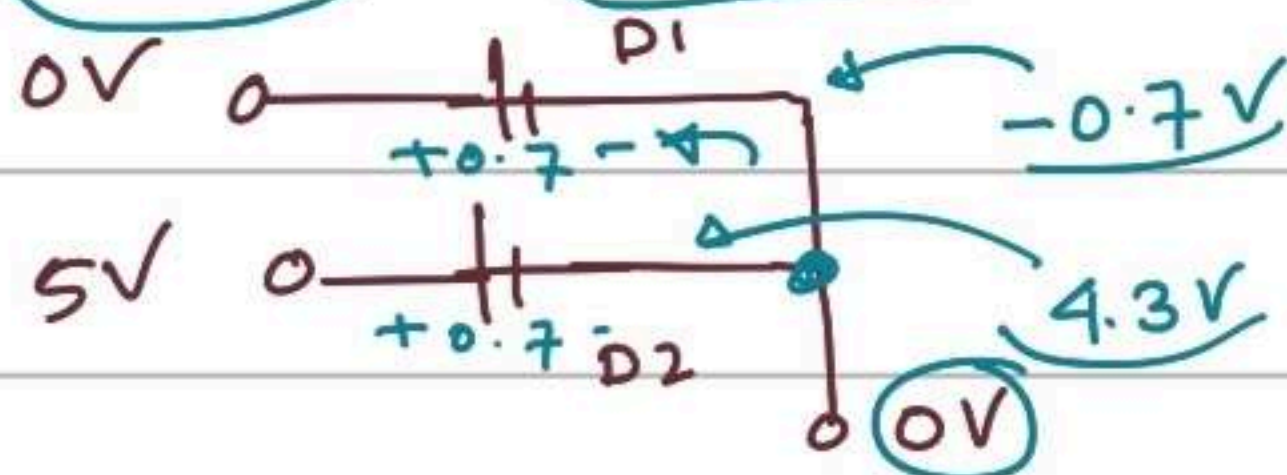
$$V_{A'} = 0V, V_{B'} = 5V$$

$$V_o = 0V$$


Contradiction.

* D1 ON, D2 OFF. # Contradiction.

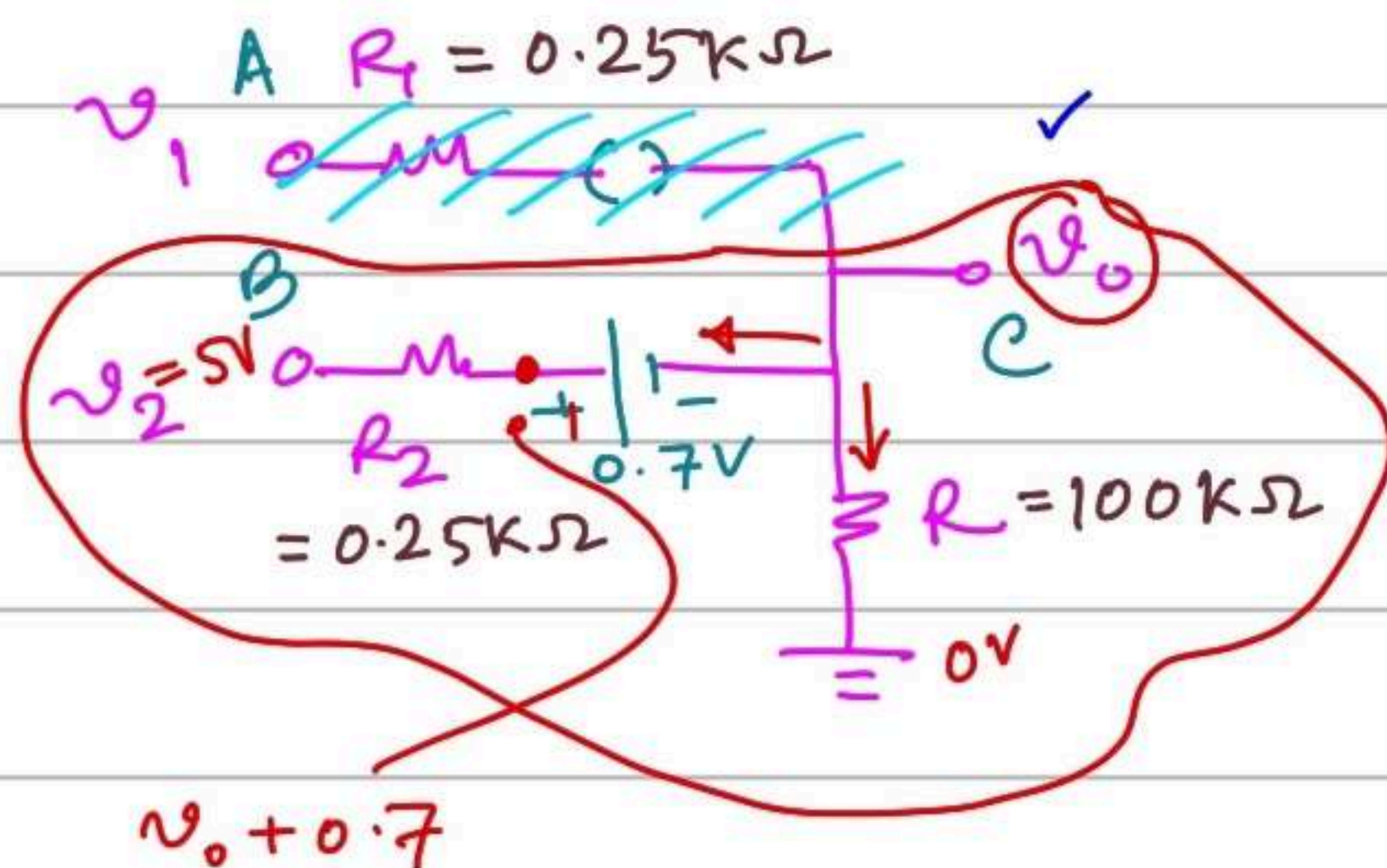
* D1 ON, D2 ON → Simplification



One node always has a single voltage.

If we do thorough analysis, we would find a contradictory result that current would flow in opposite direction through D1 diode.

* D1 OFF, D2 ON

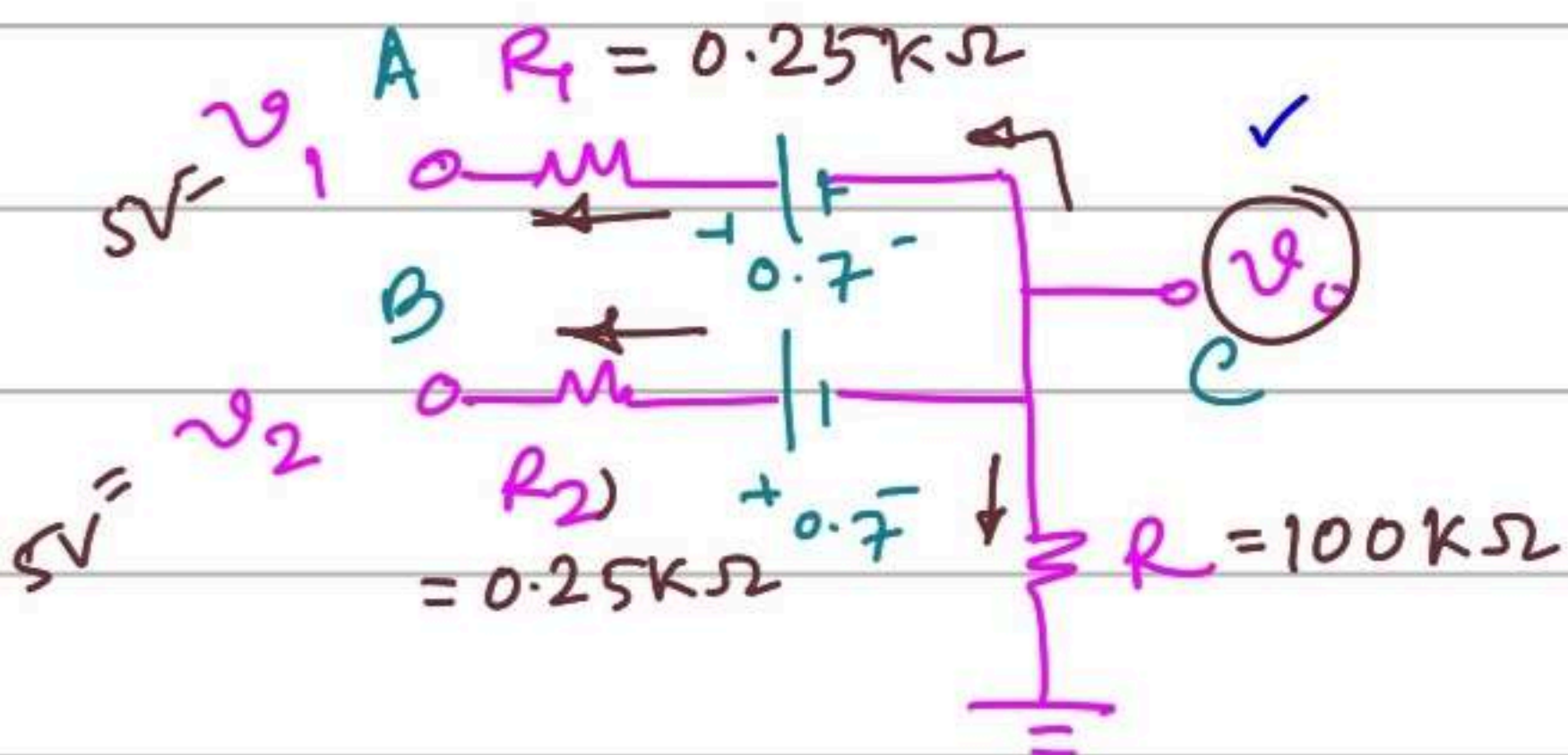


$$\frac{V_o - 0}{100K} + \frac{(V_o + 0.7) - 5V}{0.25K} = 0$$

$$\Rightarrow V_o = 4.2893V$$

Case 4: $V_1 = 5V, V_2 = 5V$

* D1 ON, D2 ON.



$$\frac{v_0 - 0}{100k} + \frac{(v_0 + 0.7) - 5V}{0.25k} \times 2 = 0$$

$$v_0 = 4.2946V$$

4.289, 4.294 We will choose the smallest of the high voltage as output voltage high.
 $v_{OH} = 4.289V \uparrow$

Power Dissipation:- Maximum dissipation of power among four cases.

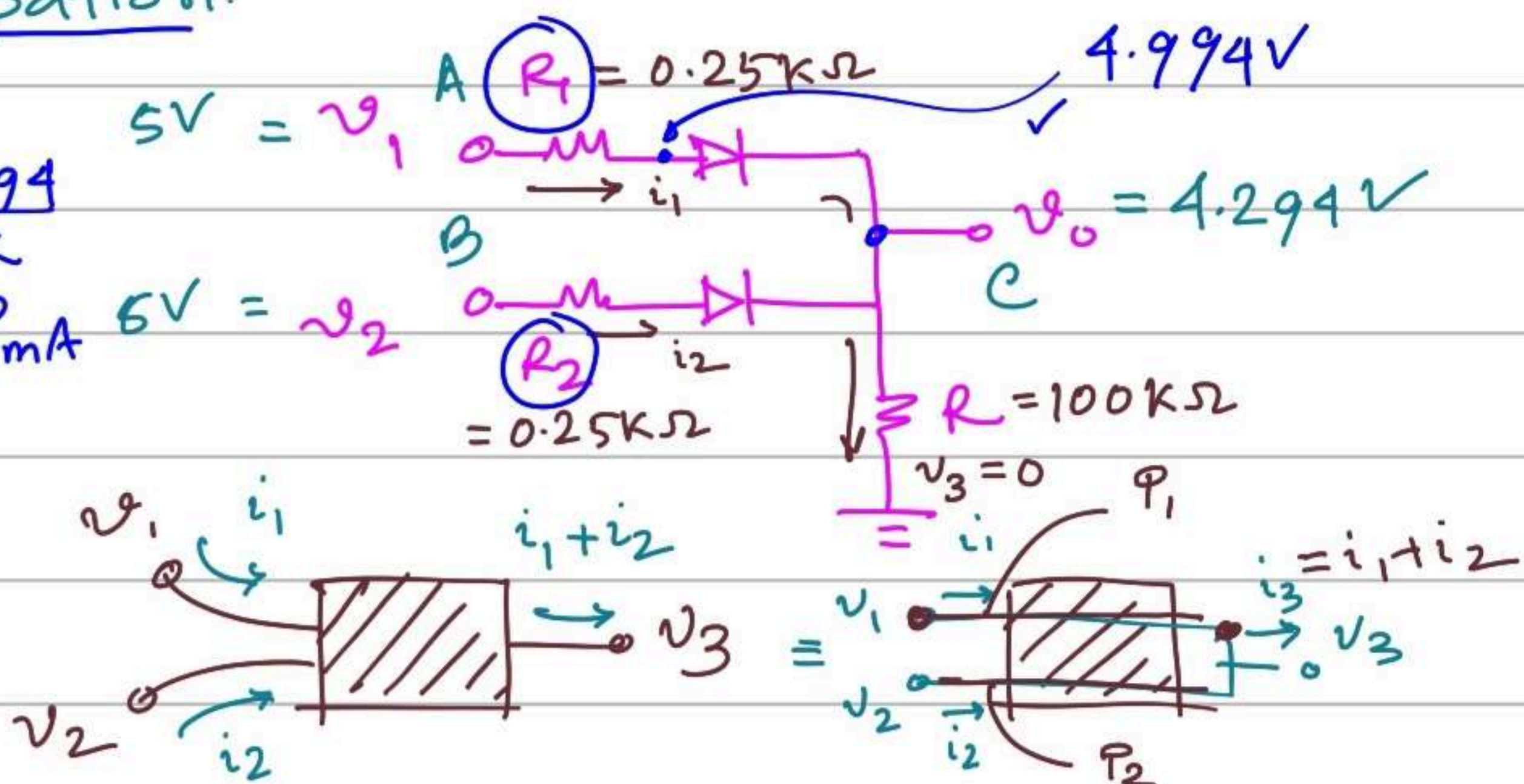
Case 4: Each individual branch is conducting current. Thus it has maximum power dissipation.

$$i_1 = \frac{5 - 4.994}{0.25k}$$

$$= 0.0215mA$$

$$i_1 = i_2$$

Theory:



$$P_1 = (v_1 - v_3) i_1, P_2 = (v_2 - v_3) i_2$$

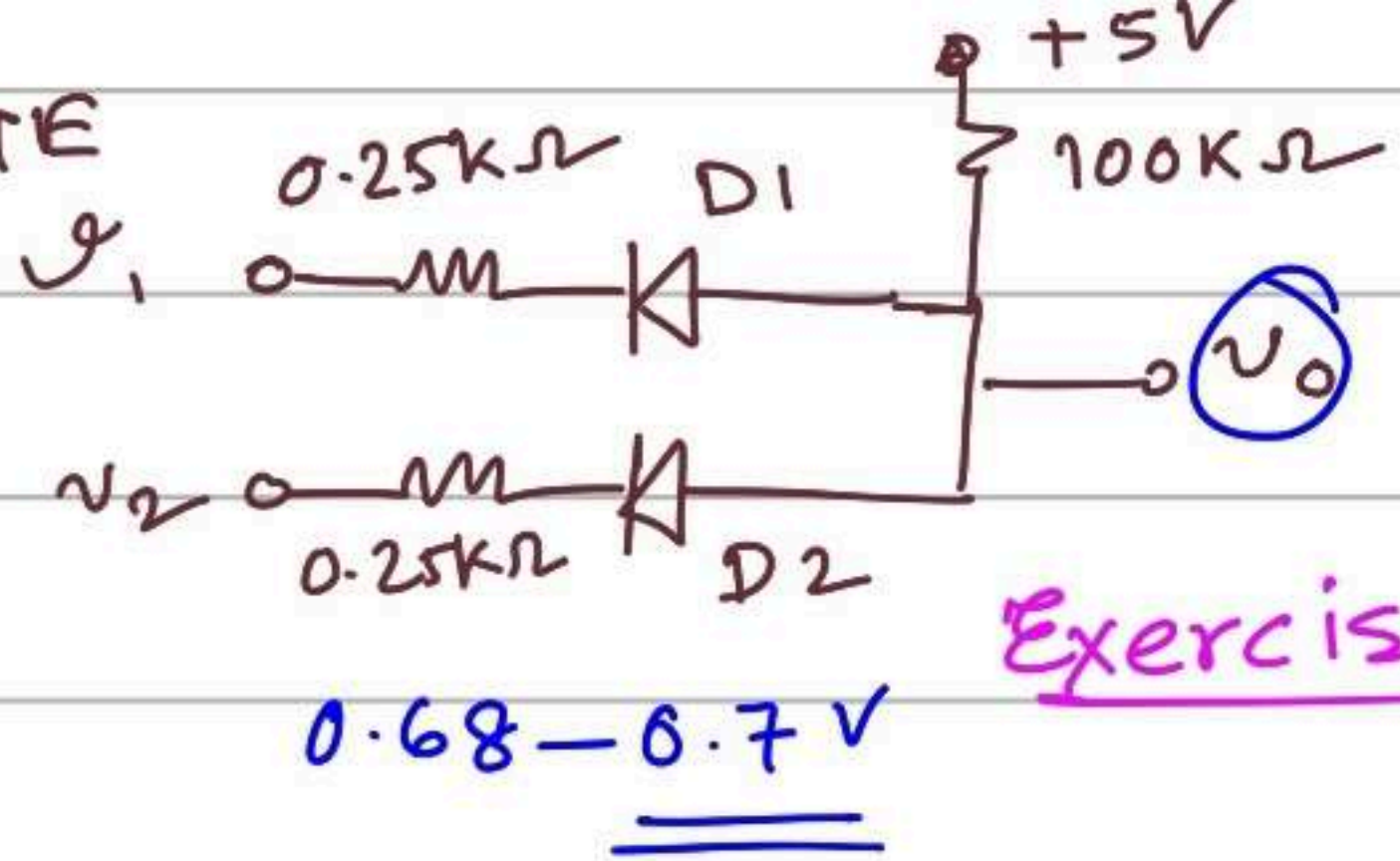
$$\text{Total power, } P = P_1 + P_2$$

$$P_2 = P_1 = (5 - 0)V(0.0215mA) \Rightarrow P = 0.215mW$$

$$V, \mu A = \mu W$$

AND GATE: 2-AND GATE

v_1	v_2	v_o	D1	D2
5V	5V	5V	OFF	OFF
5V	0V	✓	OFF	ON
0V	5V	✓	ON	OFF
0V	0V	✗	ON	ON



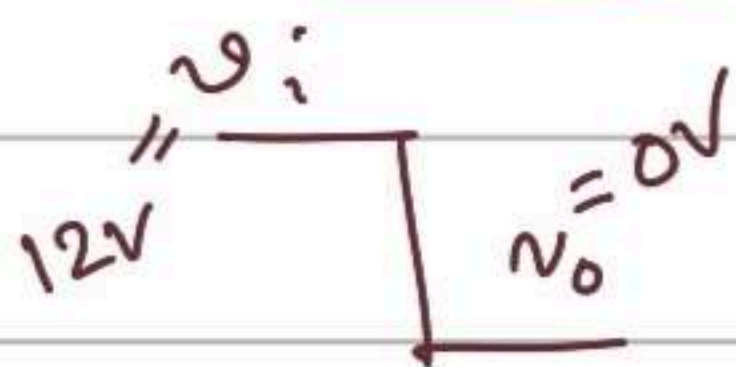
Exercise

For choosing output logical low, we will take max of the logical output low.

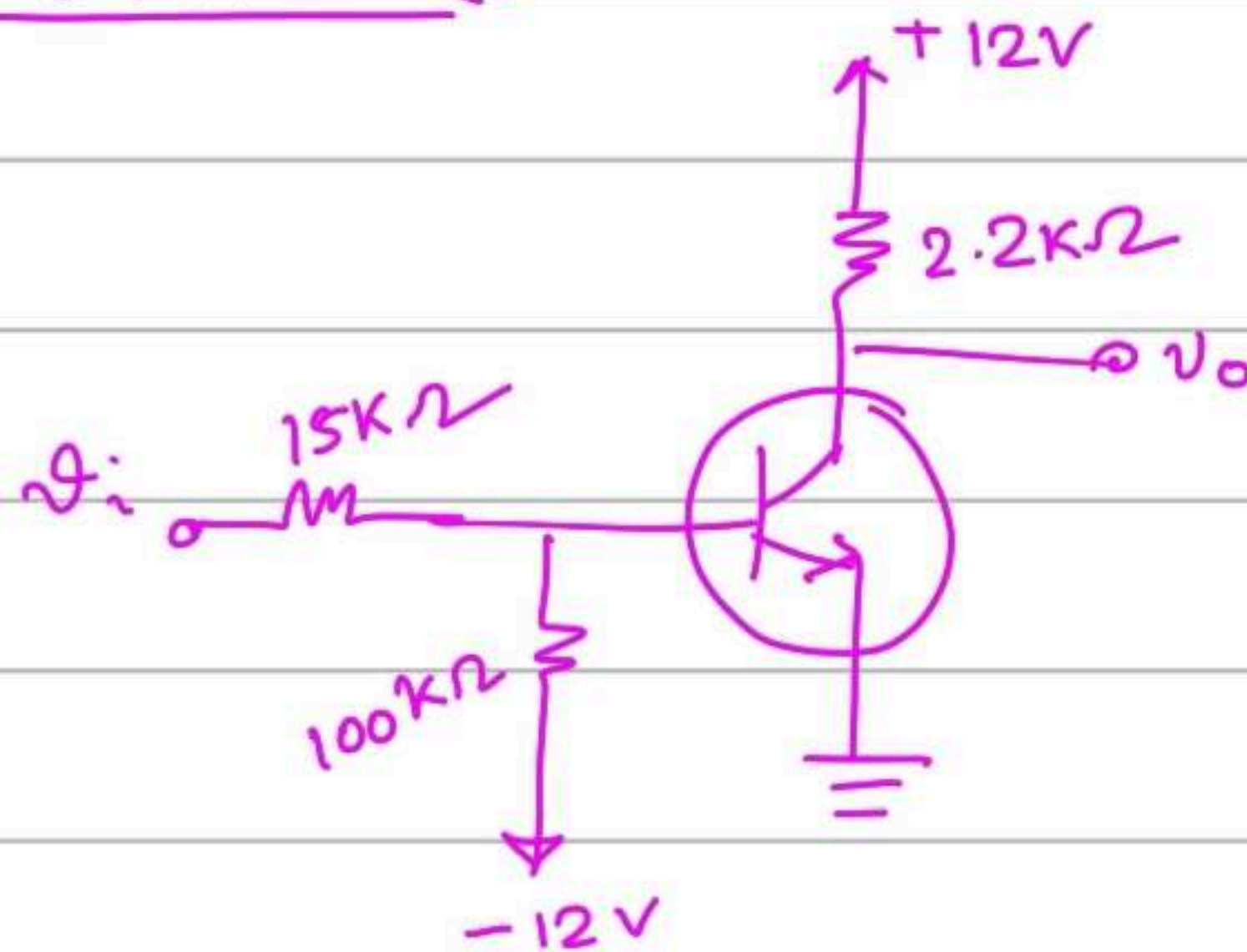
$V_{OL} = 0.7V$ $\rightarrow$ all voltages below this voltage would be considered logical low.

Register Transistor Logic (RTL)

RTL INVERTER:-



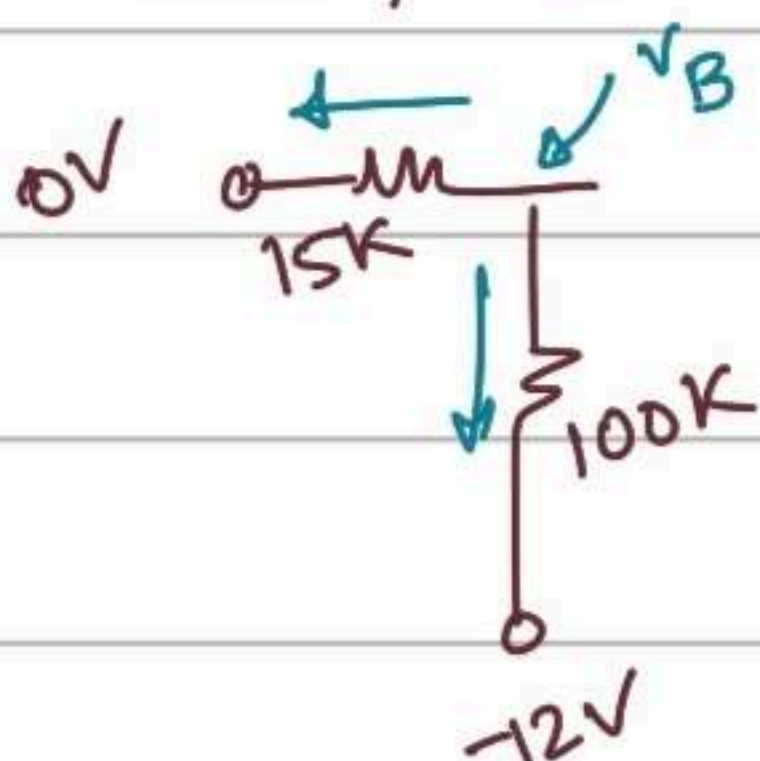
input logical
 $v_i = +12V$
low = 0V



v_i	v_o
1	0
0	1

Case 1: $v_i = 0V$

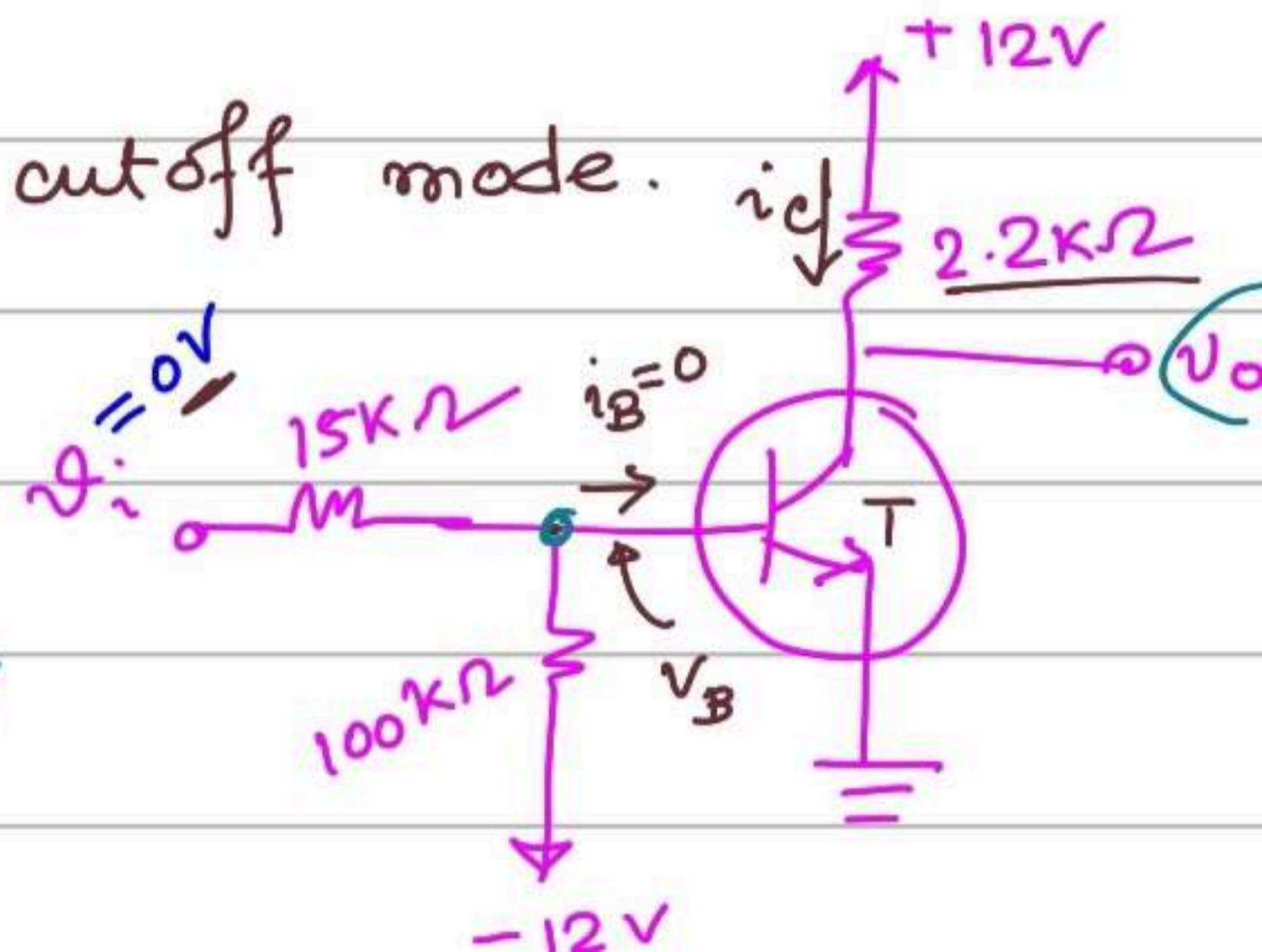
Guess/Assumption: T is in cutoff mode.



$$\frac{v_B - (-12)}{100k} + \frac{v_B - 0}{15k} = 0$$

$$\Rightarrow v_B = -1.565V$$

$$v_E = 0V$$



Verification: $V_{\gamma} = V_{BE}$ (cutin voltage) $\leq 0.5V$

$$V_{BE} = -1.565V < 0.5V \quad (\text{Our assumption is valid})$$

For cutoff mode. $i_C = i_B = i_E = 0mA$

$$i_C = \frac{12 - v_o}{2.2K} = 0 \Rightarrow \boxed{v_o = 12V} \quad \text{logical high output voltage}$$

Case ②: $v_i = +12V$

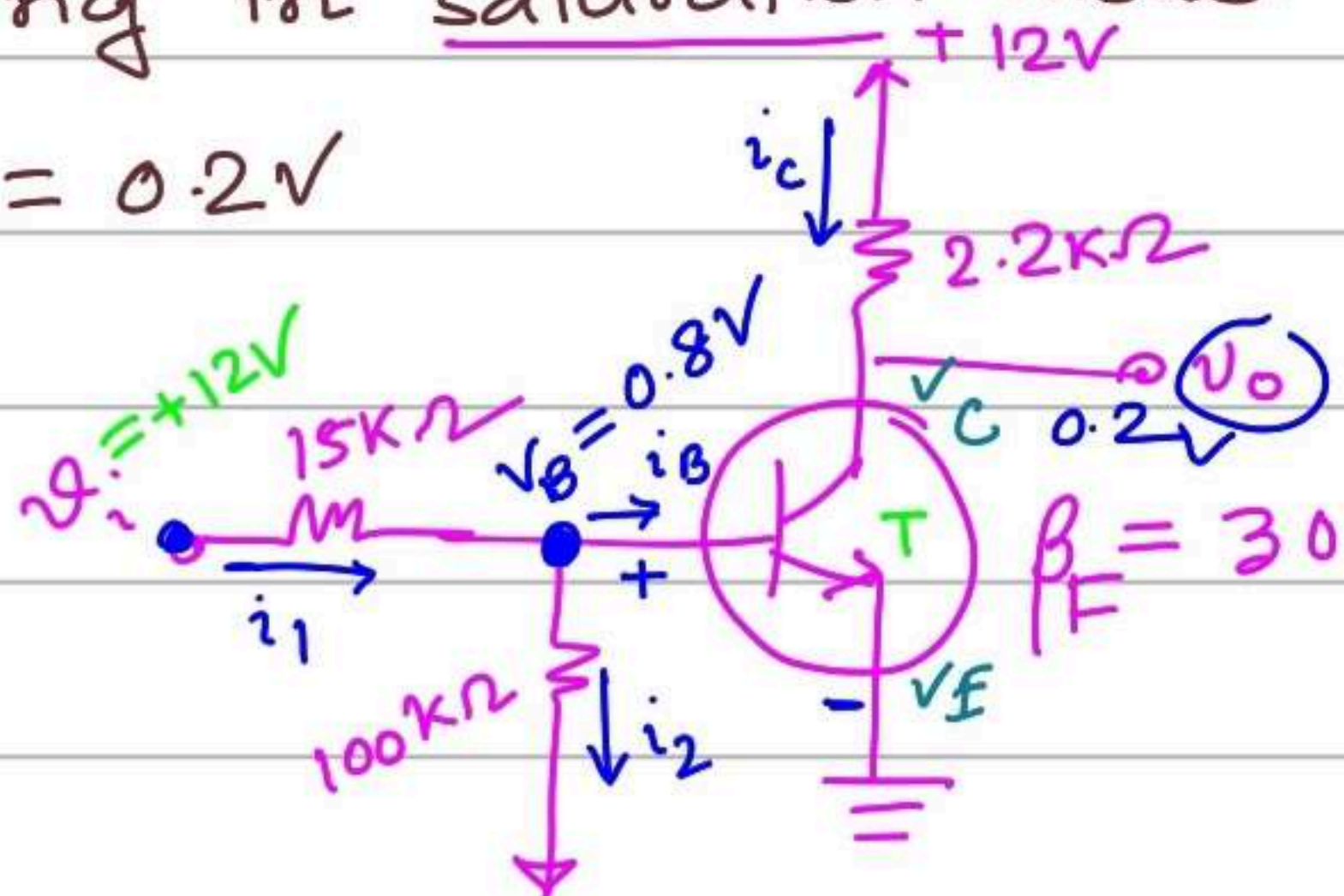
Assumption: T is operating in saturation mode.

$$\star v_o = v_C = v_C - v_E = v_{CE} = 0.2V$$

$$\star i_C = \frac{12 - 0.2}{2.2K} = 5.36mA$$

$$\star i_1 = \frac{12 - 0.8}{15K} = 0.746mA$$

$$\star i_2 = \frac{0.8 - (-12)}{100K} = 0.128mA$$



When transistor is operating in saturation mode $V_{BE} = 0.8V$, $V_{CE} = 0.2V/0.1V$

$$i_B = i_1 - i_2 = 0.619mA$$

Verification: $\beta_{\text{forced}} = \frac{i_C}{i_B} = 8.659$

Our assumption is valid.

$$< 30 = \beta_F$$

Thus our analysis shows output voltage

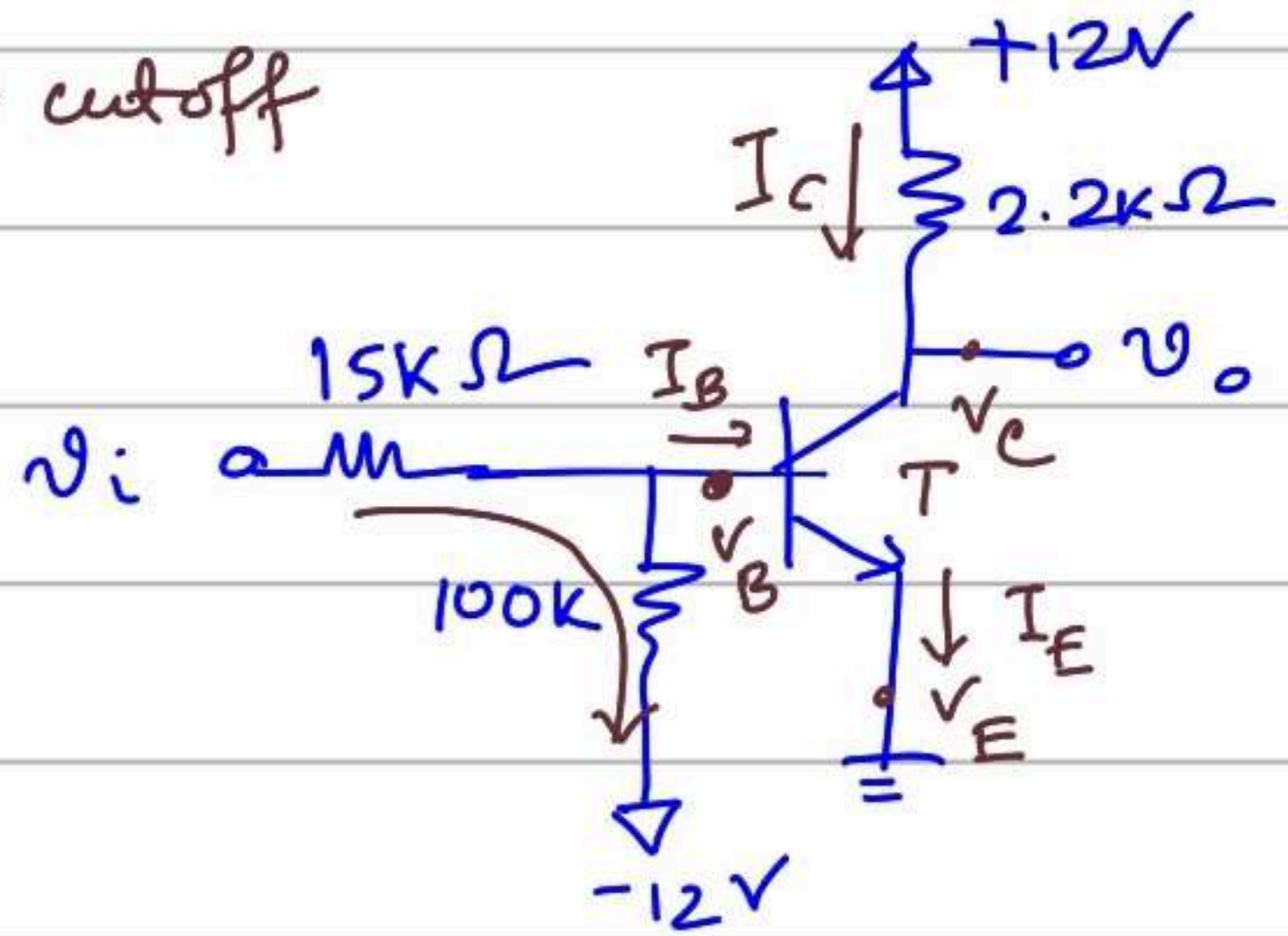
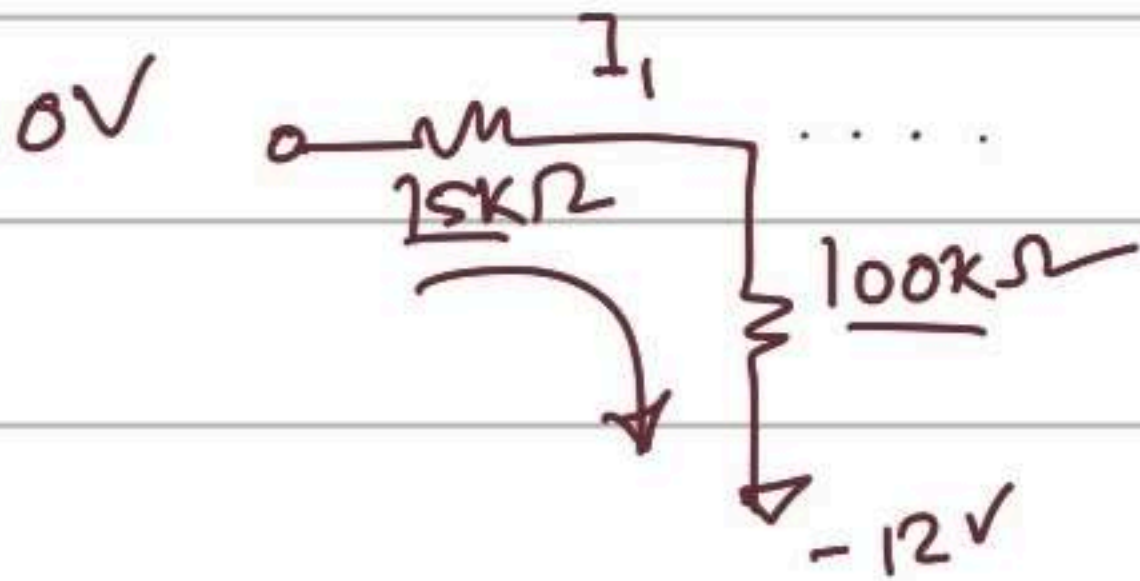
is now $0.2V$. = logical low voltage at the output terminal.
 $\boxed{V_{OL} = 0.2V}$

Lecture 4: RTL circuits : Power dissipation, Noise Margin and Fanout Calculation.

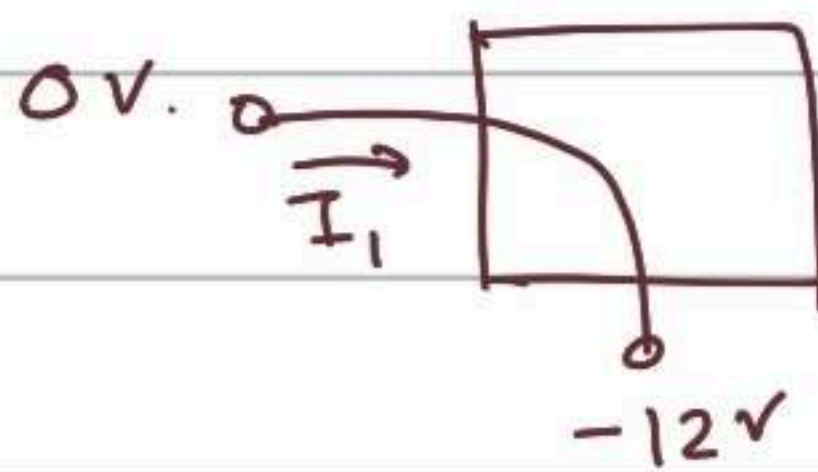
Power dissipation in RTL inverter :-

Case ①. $v_i = 0V$. $T \rightarrow$ cutoff

$$I_B = I_C = I_E = 0mA$$



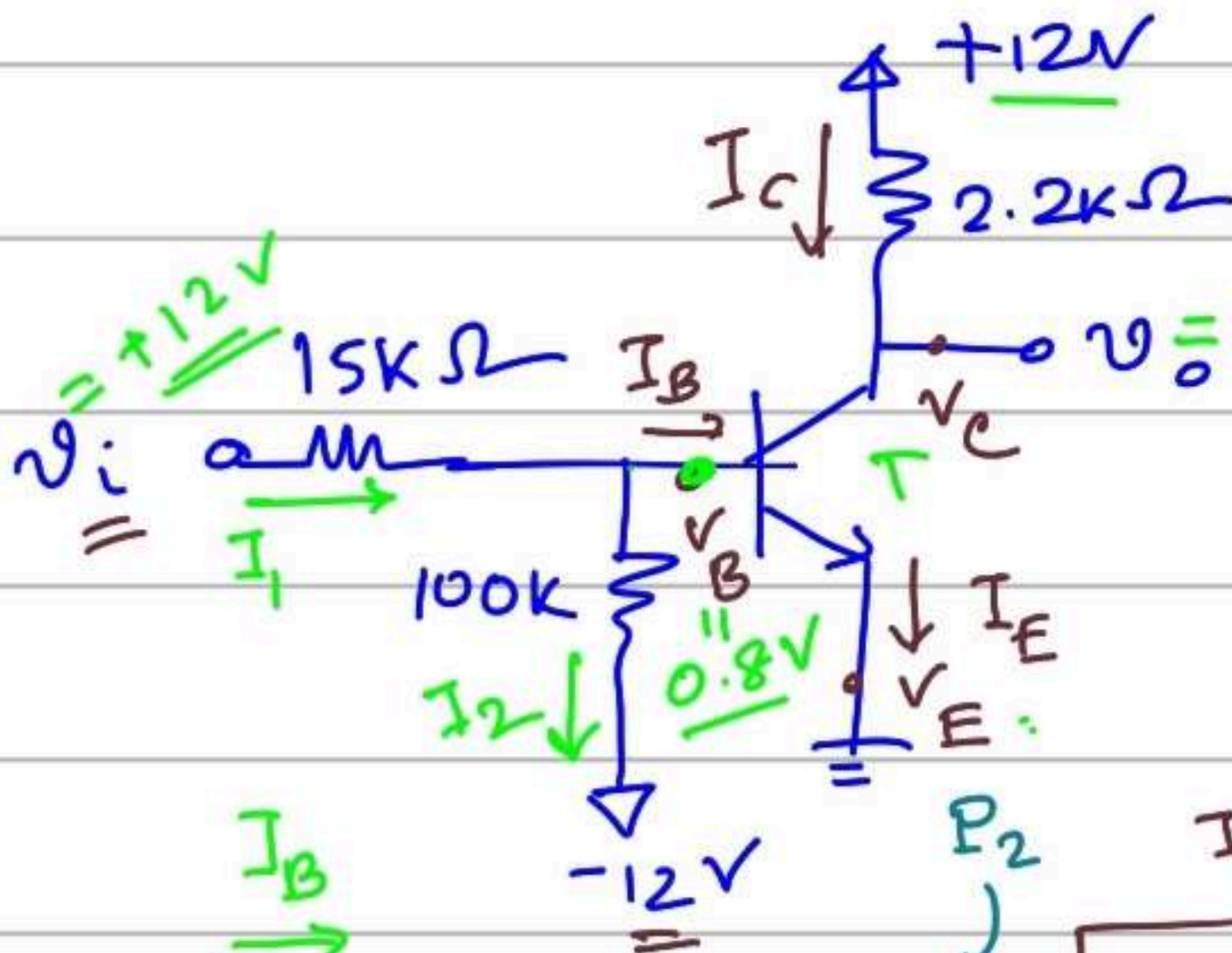
$$I_1 = \frac{0 - (-12)}{115k\Omega} = 0.104mA$$



$$P = (0 - (-12)) \times 0.104 = \boxed{1.2521mW}$$

Case ② $v_i = +12V$. $T \rightarrow$ saturation (RTL)

$$V_{BE} = 0.8V, V_{CE} = 0.2V \text{ (0.1V)}$$

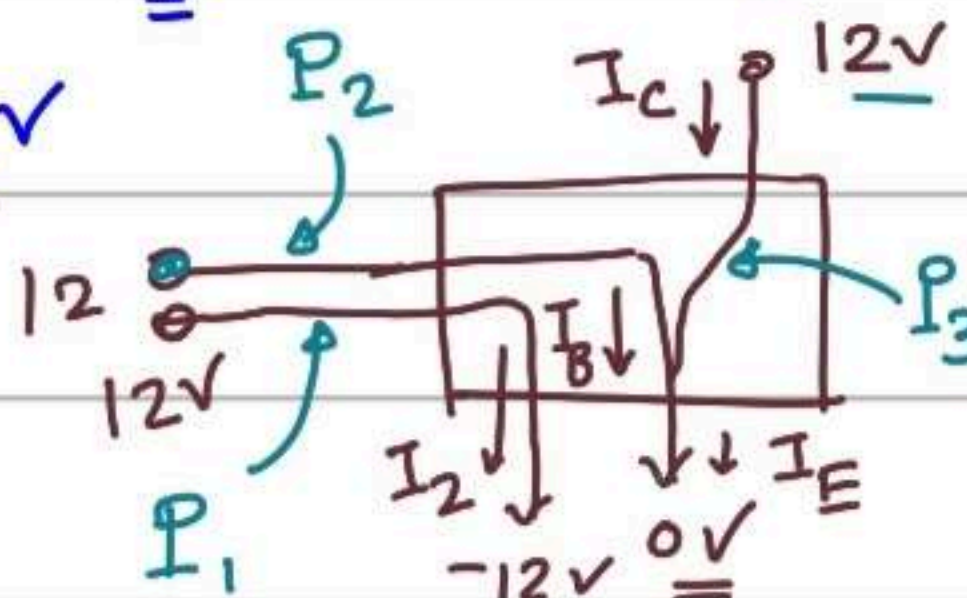
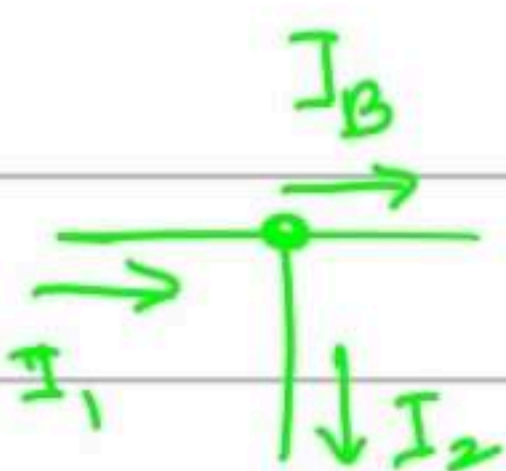


$$I_C = \frac{12 - 0.2}{2.2K} = 5.3636mA$$

$$I_1 = \frac{12 - 0.8}{15K} = 0.7466mA$$

$$I_2 = \frac{0.8 - (-12)}{100K} = 0.128mA$$

$$I_B = I_1 - I_2 = 0.6186mA$$



$$P_1 = (12 - (-12)) \times 0.128 = 3.072mW$$

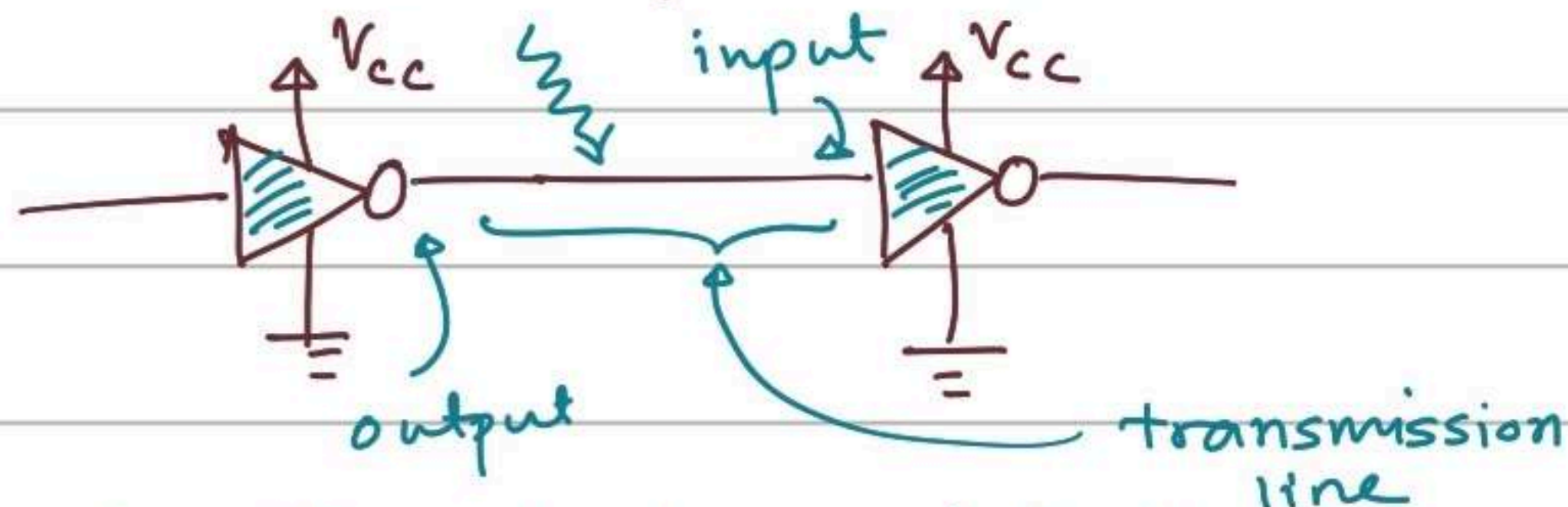
$$P_2 = (12 - 0) \times 0.6186 = 7.4323mW$$

$$P_3 = (12 - 0) \times 5.3636 = 64.3632 \text{ mW}$$

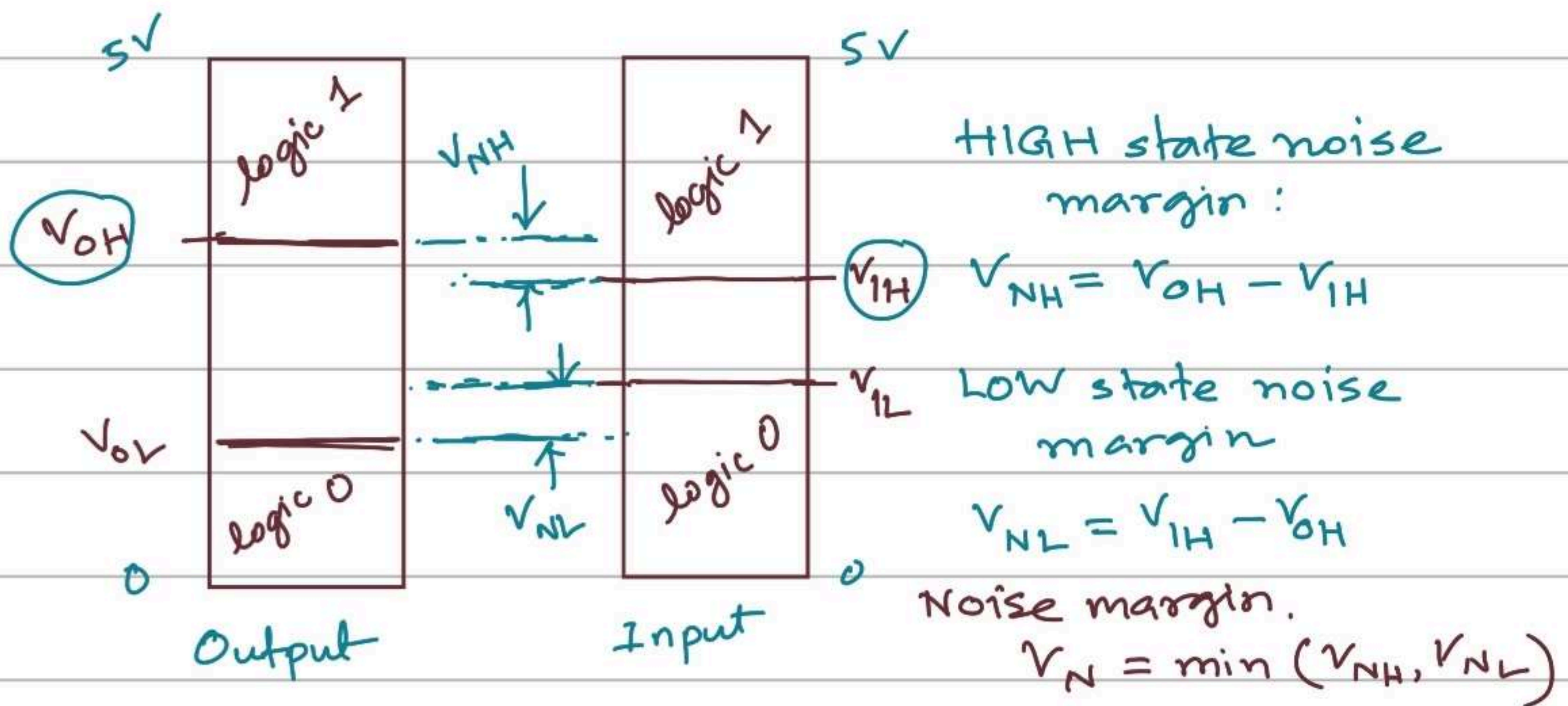
$$\text{Total power dissipation, } P = P_1 + P_2 + P_3 = \boxed{74.8534 \text{ mW}}$$

NOISE MARGIN:-

The amount of noise voltage a ^{digital logic} circuit can tolerate while successful transmission.



We model different types interference or noises using "noise voltage."



Why do we use minimum?

→ To ensure safety margins for both high and low voltages.

$\frac{V_{OH}}{V_{IH}}$ → minimum voltage level at an output in the logical "1" state under defined load condition.

$V_{OL} \rightarrow$ maximum " " at an output in the logical "0" state under defined ^{input} load condition.

Example: Output voltage is low when T is in saturation. $V_{OL} = 0.2V$ (Saturated logic family)

* $V_{OH} = ?$ (This will be given)

The output ^{high} voltage can be allowed to decrease 0.5V from its maximum value.

* $V_{OH} = +12 - 0.5V = 11.5V$

* V_{IL} calculation:

* When input voltage is low, transistor operates in cutoff mode.

* For RTL circuit, V_{IL} should be equal to the maximum voltage that takes the transistor on the verge of turn off to turn on condition.

* The transistor will turn on if $V_{BE} \geq 0.5V$.

* We need find the input voltage so that

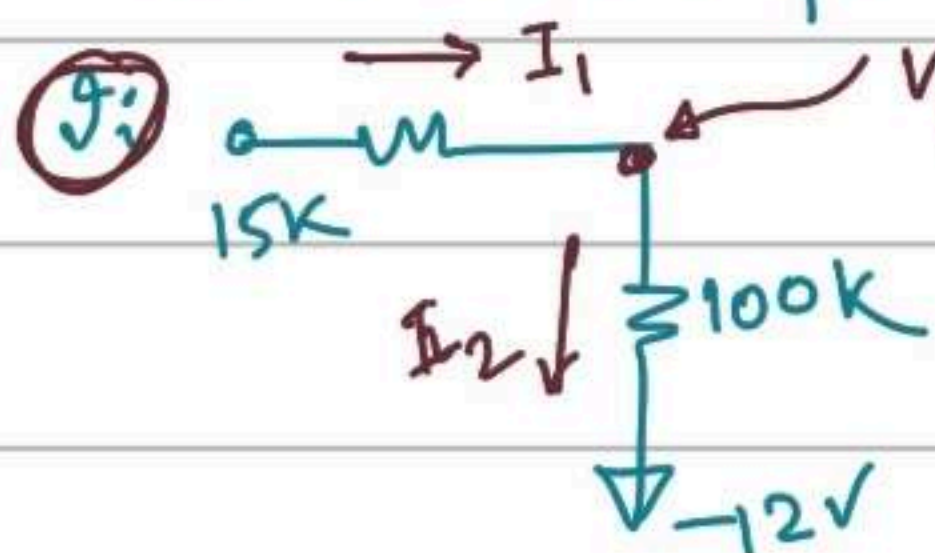
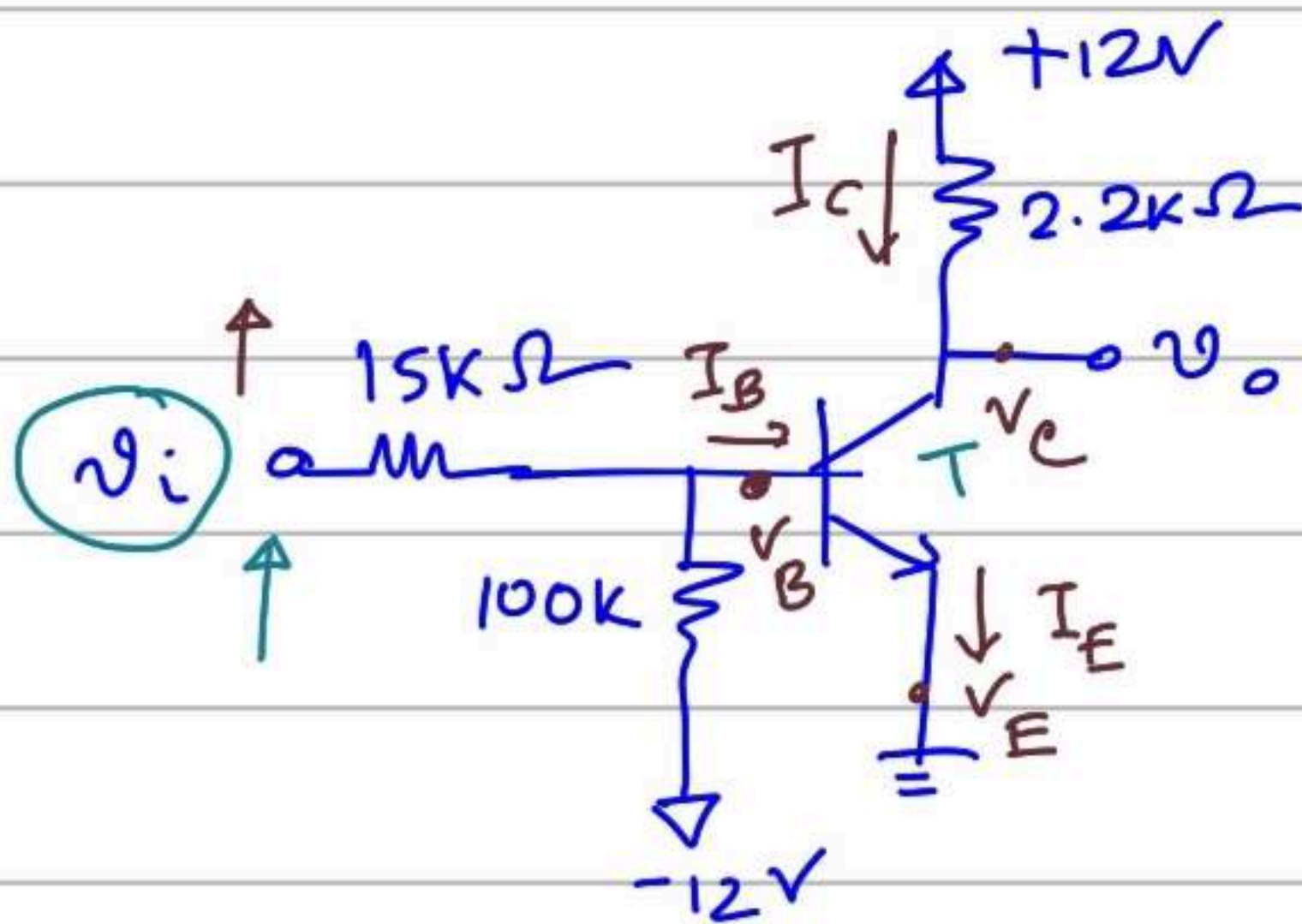
$V_{BE} = 0.5V$ but we assume transistor is in cutoff.

* T $\rightarrow$ cutoff, $I_B = 0mA$ $I_2 = \frac{0.5 - (-12)}{100k} = 0.125mA$

$I_1 = I_2$, $I_1 = \frac{V_i - 0.5}{15k} = 0.125mA$

$\therefore V_i = 0.5 + 15 \times 0.125 = 2.375V$

$V_{IL} = 2.375V$



V_{IH} calculation: When input voltage is high, transistor

T operates in saturation mode. $V_{CE} = 0.2V = v_o$.

* The minimum voltage at the input terminal which will change the operating mode of transistor

T to saturation to forward active mode should be V_{IH} .

* We will assume T is saturation,

but $\beta_{forced} \approx \beta_F$. [Transition point between sat. and F.A.]

* T $\rightarrow$ saturation. $I_C = \frac{12 - 0.2}{2.2k} = 5.3636 \text{ mA}$

$$\beta_{forced} = \frac{I_C}{I_B} = 30 \Rightarrow I_B = \frac{I_C}{\beta} = 0.1788 \text{ mA}.$$

$$I_2 = \frac{0.8 - (-12)}{100k} = 0.128 \text{ mA}. \quad I_1 = I_2 + I_B = 0.3068 \text{ mA}$$

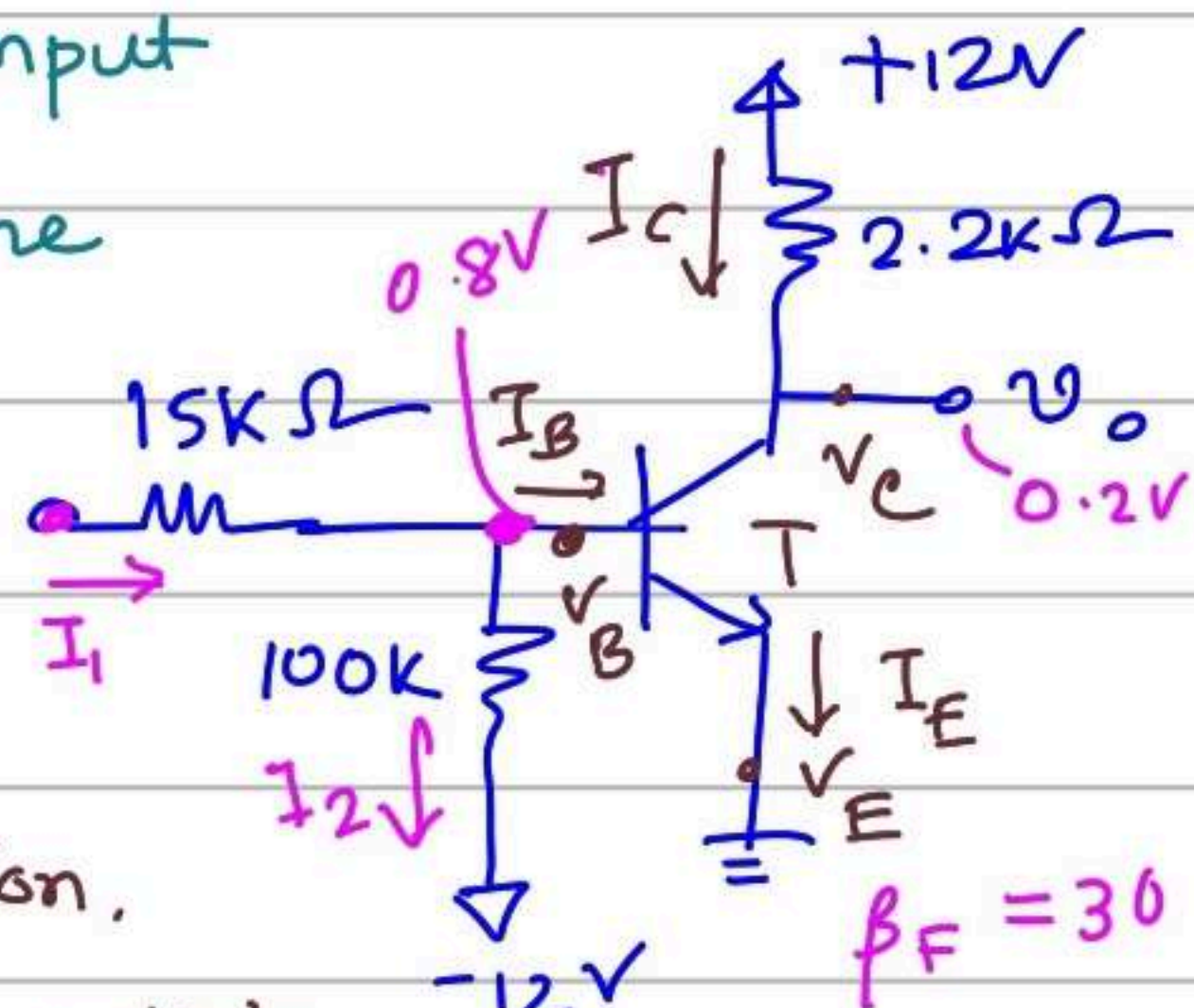
$$\frac{v_i - 0.8}{15k} = 0.3068 \text{ mA} \Rightarrow v_i = 0.8 + 15 \times 0.3068 = 5.4018 \text{ V}$$

$$\boxed{V_{IH} = 5.4018 \text{ V}}$$

$$\text{High state noise margin: } V_{NH} = V_{OH} - V_{IH} = 6.0982 \text{ V}$$

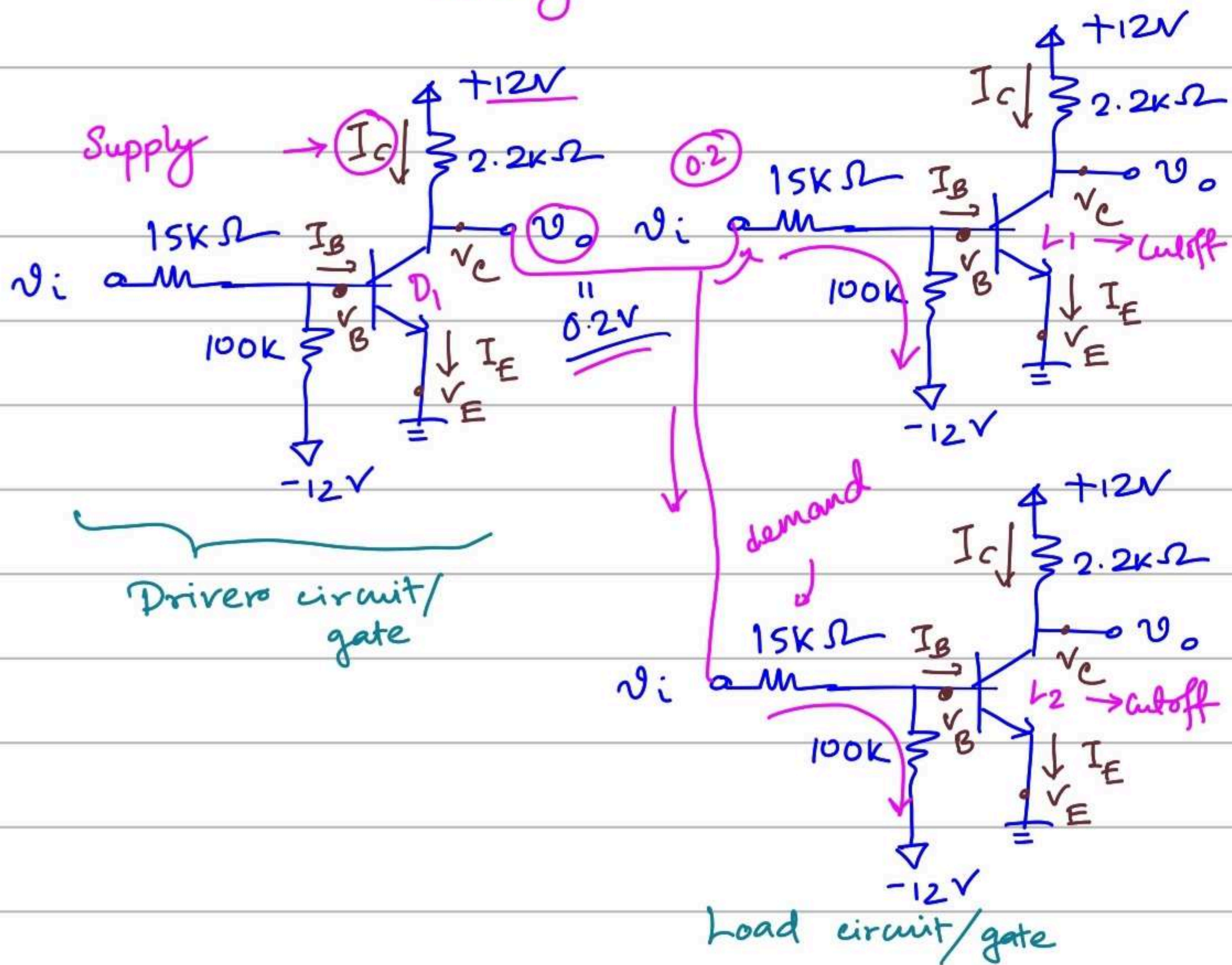
$$\text{Low state noise margin: } V_{NL} = V_{IL} - V_{OL} = 2.175 \text{ V}$$

$$\text{Noise margin} = \min(6.0982, 2.175) = \boxed{2.175 \text{ V}}$$



Fanout: The number of input terminals are connected to a output terminal.

Maximum fanout (fanout): The maximum number of logic inputs of the logic family that an output terminal can drive, is called maximum fanout.
reliably

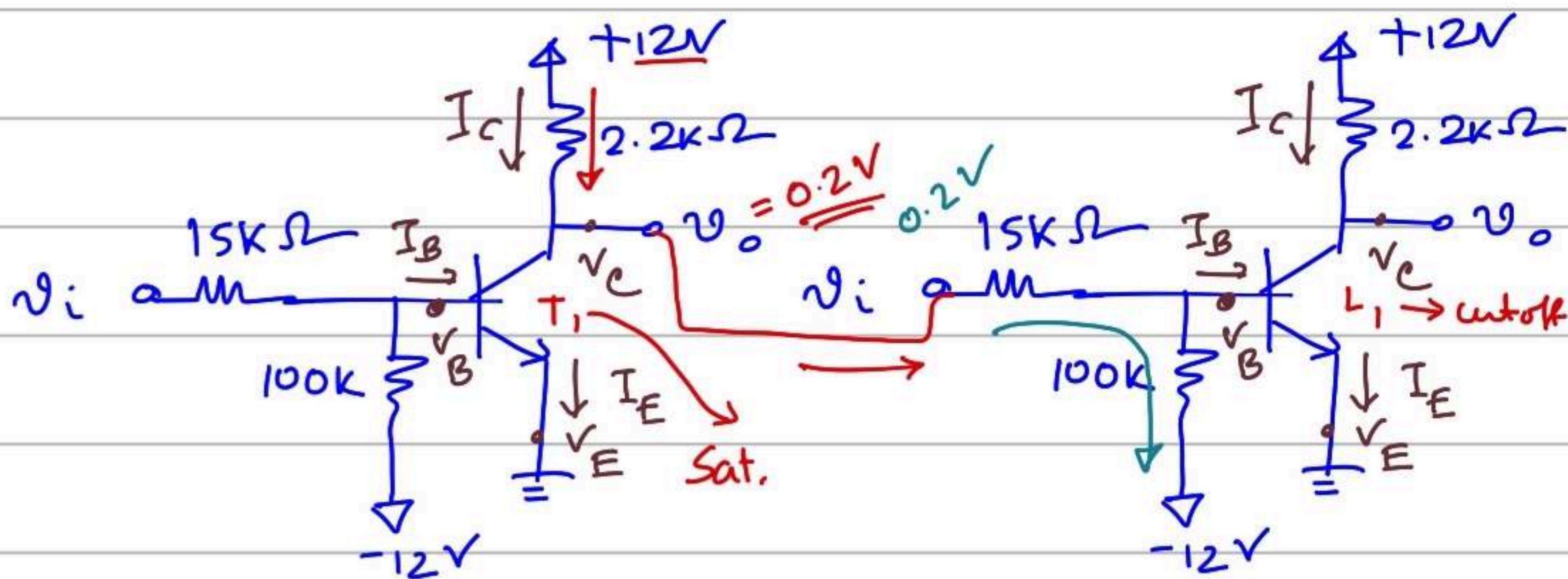


Question: Calculate the maximum fanout of the above RTL circuit. Assume $V_{OH} = 10V$.

Case ①: The output of the driver circuit is low.

* Maximum supply current :- The maximum of current can be delivered by the driver circuit without changing its operating mode.

* Individual demand load current :- The maximum amount of current needed by each individual load while driven by driver circuit.

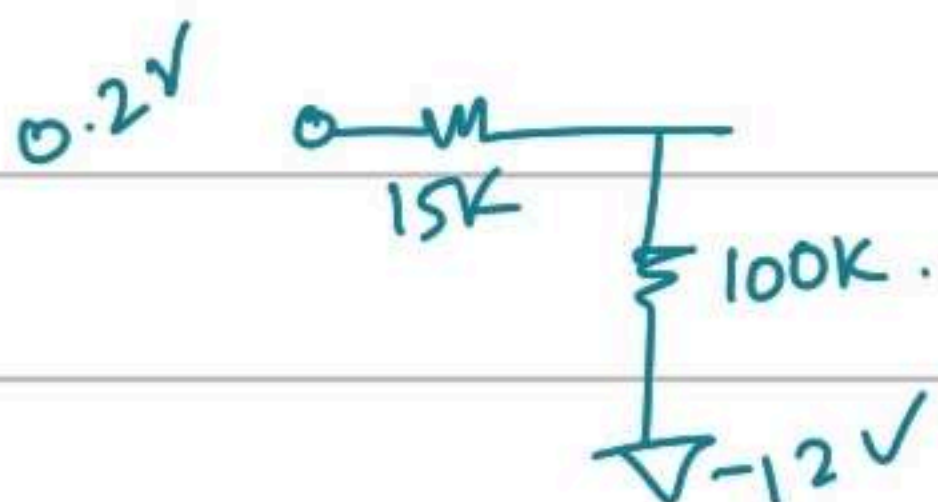


Driver circuit $T_1 \rightarrow$ Saturation. $V_0 = 0.2V$

Load circuits $L_1, L_2 \rightarrow$ cutoff. $I_B = 0$.

* Maximum supply current, $\frac{12-0.2}{2.2k} = 5.3636 \text{ mA}$

* Individual demand load current, $\frac{0.2-(-12)}{(15+100)k} = 0.106 \text{ mA}$



maximum fanout for this case,

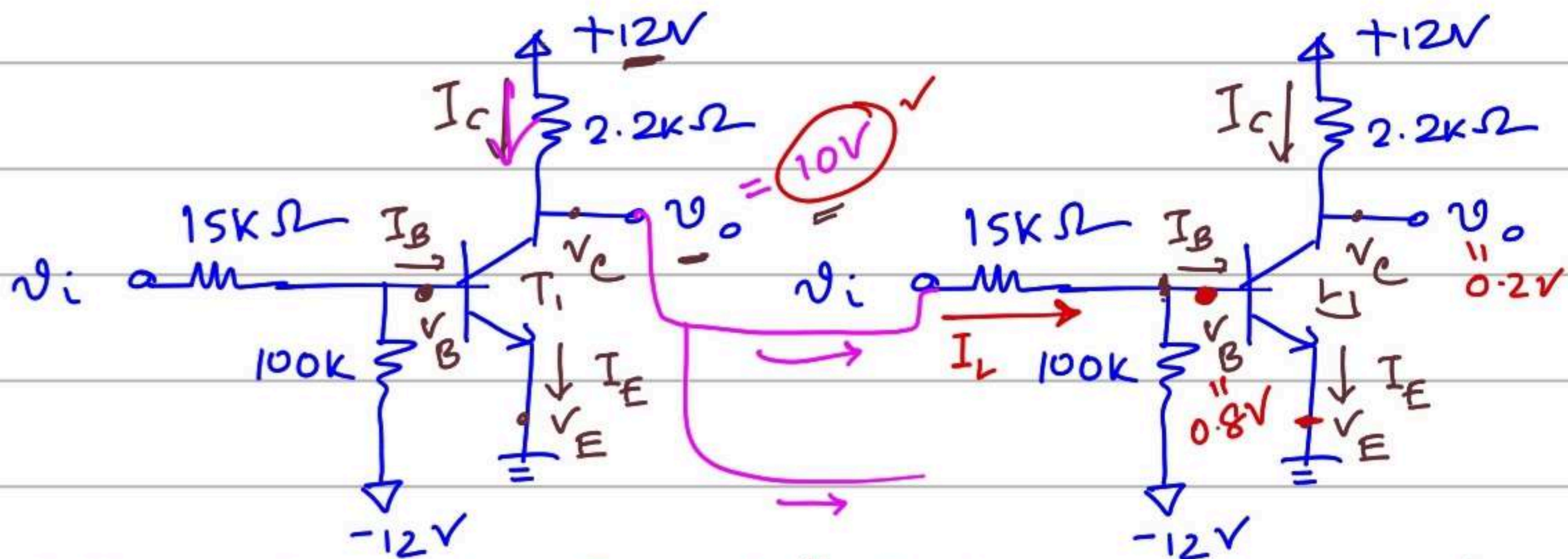
$$= \left\lfloor \frac{5.3636}{0.106} \right\rfloor = \left\lfloor 50.6 \right\rfloor = 50$$

* If we connect 51 load circuits, then demand exceeds supply. Thus driver circuit will malfunction.

Case ② : The output voltage of the driver circuit is High.

* If we connect load circuits when output of driver is high, the output voltage of driver starts to drop.

$$V_{OH} = 10V$$



* Driver transistor, $T_1 \rightarrow$ cutoff, load circuit's transistors $T_2 \rightarrow$ Sat.

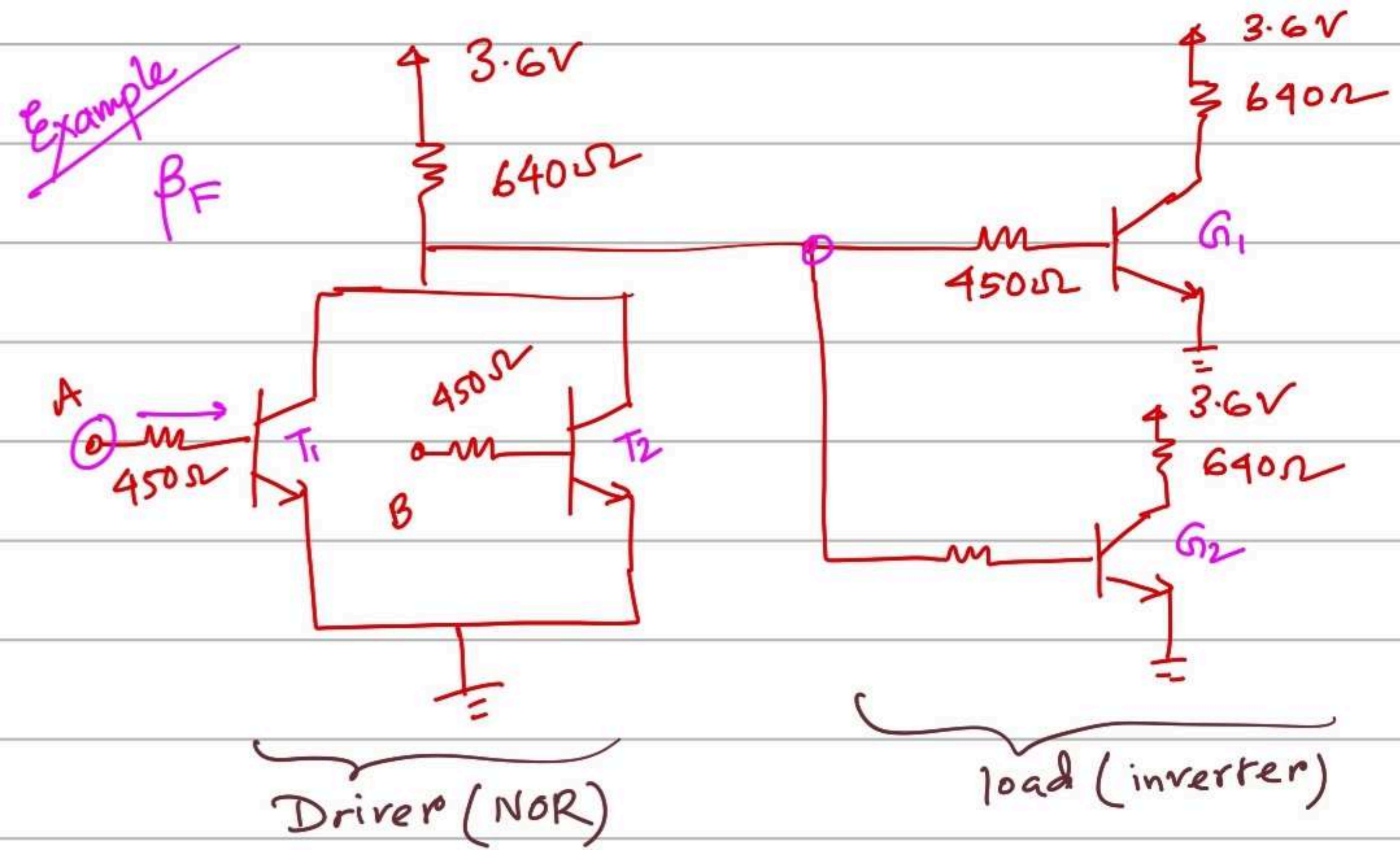
* Maximum supply current = $\frac{12-10}{2.2k\Omega} = 0.909mA$

* Individual load current, $I_L = \frac{10-0.8}{15k} = 0.6133mA$

* Maximum fanout for this case = $\left\lfloor \frac{0.909}{0.6133} \right\rfloor = \left\lfloor 1.48 \right\rfloor = 1$

⇒ Overall maximum fanout must be calculated by considering worst case scenario. To prevent malfunction of driver circuit, we need to take the minimum of the two cases.

⇒ Maximum fanout, = $\min(50, 1) = 1$



* There would be resistors connected to input terminal and the resistor is connected to the transistor. (RTL circuit)

(RTL)
 * If the input ^{high} voltage, you can assume ^{switching} transistor is operating in saturation, or if the input voltage is low, transistor is operating in cutoff mode.

* Question: If the number of fanout is 5. Find the minimum value of β_F , so that these circuits operate as NOR gate. Assume all transistors have same value of β .